

ECP4.2.2 Project Work And Dissertation

Design of ML Assisted Beat to Beat Blood Pressure Estimator and It's Related Hardware Implementation

By

Swapnil Saha

Reg No: 012-1111-0753-20

Roll No: T91/ECE/226013

B. Tech, 8th Semester, ECE

Ritam Pal

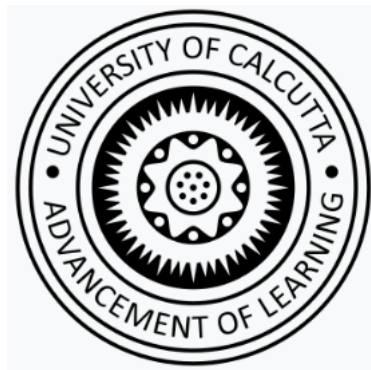
Reg No: 425-1114-0868-19

Roll No: T91/ECE/216007

B. Tech, 8th Semester, ECE

Under The Guidance
Dr. Sumitra Mukhopadhyay

Institute of Radiophysics and Electronics



UNIVERSITY OF CALCUTTA

UNIVERSITY



OF CALCUTTA

UNIVERSITY COLLEGE OF SCIENCE AND TECHNOLOGY
INSTITUTE OF RADIO PHYSICS & ELECTRONICS

Telephone : (+91) (33) 2350 9115/9116/9413
Web : www.irpel.in

SISIR MITRA BHAVAN
92, Acharya Prafulla Chandra Road
Kolkata - 700 009, India

Certificate of Supervisor

This is to certify that Swapnil Saha, Registration no. 012–1111–0753–20, Roll no. T91/ECE/226013 and Ritam Pal, Registration no. 425–1114–0868–19, Roll no. T91/ECE/216007 have carried out their B.Tech., Eighth Semester, ECP4.2.2 Project work and Dissertation entitled “Design of ML Assisted Beat to Beat Blood Pressure Estimator and It’s Related Hardware Implementation” in partial fulfilment of requirement for the degree of Bachelor of Technology in Electronics and Communication Engineering, University of Calcutta, under my supervision.

(Sumitra Mukhopadhyay)
(Project Supervisor)

Place: Kolkata

Date :

UNIVERSITY



OF CALCUTTA

UNIVERSITY COLLEGE OF SCIENCE AND TECHNOLOGY
INSTITUTE OF RADIO PHYSICS & ELECTRONICS

Telephone : (+91) (33) 2350 9115/9116/9413
Web : www.irpel.in

SISIR MITRA BHAVAN
92, Acharya Prafulla Chandra Road
Kolkata - 700 009, India

Certificate of Approval

The foregoing project is thereby approved as a creditable study of a Bachelor of Technology in Electronics and Communication Engineering and presented in a manner satisfactory to warrant acceptance as pre-requisite to the degree for which it has been submitted. It is understood that by this approval, the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein but approve the project report only for which it is submitted.

Committee on final examination
for the evaluation of project report

Examiners

Declaration of Originality and Compliance of Academic Ethics

We hereby declare that this project report contains a combination of literature survey and original research work carried out by the undersigned candidate(s) as part of the Bachelor of Technology (B.Tech) in Electronics and Communication Engineering.

All information presented in this document has been obtained and compiled in accordance with established academic rules and ethical standards. We further declare that all sources of information, including data, figures, and results that are not original to this work, have been properly cited and acknowledged.

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Furthermore, we declare that no generative AI tools or AI-assisted technologies have been used in the writing or preparation of this project report.

Name: Swapnil Saha

Registration: 012-1111-0753-20

Roll No.: T91/ECE/226013

Name: Ritam Pal

Registration: 425-1114-0868-19

Roll No.: T91/ECE/216007

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(Ritam Pal)

(Swapnil Saha)

Abstract

We have designed an end-to-end system integrating biomedical signal processing, machine learning, and embedded hardware for continuous and non-invasive blood pressure monitoring was developed. The MIMIC-III Waveform Database was utilized to ensure robust model training and reliable validation across diverse physiological conditions.

The preprocessing of the Photoplethysmography (PPG) signals was performed using a 4th-order Butterworth filter, through which noise, motion artifacts, and baseline drift were effectively minimized to enhance signal quality. Then meaningful latent representations of the filtered PPG signals were then extracted using a Variational Autoencoder (VAE), enabling efficient feature learning from high-dimensional time-series data.

Subsequently, a Random Forest regression model was trained on the extracted features to accurately estimate Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP). The performance of the proposed model was rigorously evaluated using Mean Absolute Error (MAE) and Standard Deviation of Error (SDE), demonstrating reliable predictive capability and consistency.

In addition, a hardware prototype was designed and implemented for real-time physiological signal acquisition using a PPG sensor integrated with an Analog-to-Digital Converter (ADC). The complete real-time signal processing and inference pipeline was deployed on a Raspberry Pi 4 Model B, enabling continuous monitoring and on-device prediction.

This report details the development of a non-invasive Photoplethysmography (PPG) system designed to monitor cardiovascular activity in real-time. The system utilizes a TCRT1000 reflective sensor integrated with a custom-built analog front-end, featuring a Trans-impedance Amplifier (TIA) and dual-stage filtering (High Pass and Low Pass) to ensure high signal integrity. Data acquisition is performed by an ESP32 microcontroller, which transmits processed data serially to a Raspberry Pi host. The final output, representing heart rate in Beats Per Minute (BPM), is displayed locally on an I2C 16x2 LCD module. This project demonstrates a complete hardware-software pipeline for affordable health monitoring.

The proposed system demonstrates a scalable and cost-effective solution for real-time blood pressure estimation, with potential applications in wearable healthcare devices, remote patient monitoring, and next-generation biomedical instrumentation.

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CHAPTER 1

1 Introduction

1.1 Blood Pressure

Blood pressure is an important physiological signal that reflects a person's overall health. It is the pressure exerted by blood on the walls of the arteries during the different phases of the cardiac cycle. Blood pressure varies according to a person's physical and mental activities, and frequent fluctuations can be a sign of underlying health issues. Blood pressure is measured using two primary values:

- Systolic blood pressure – the maximum arterial pressure during cardiac systole (when the ventricles contract).
- Diastolic blood pressure – the minimum arterial pressure during cardiac diastole (when the heart relaxes).

Elevated blood pressure is commonly referred to as hypertension. An individual is generally considered hypertensive if their systolic blood pressure is 140 mmHg or higher, and/or their diastolic blood pressure is 90 mmHg or higher.

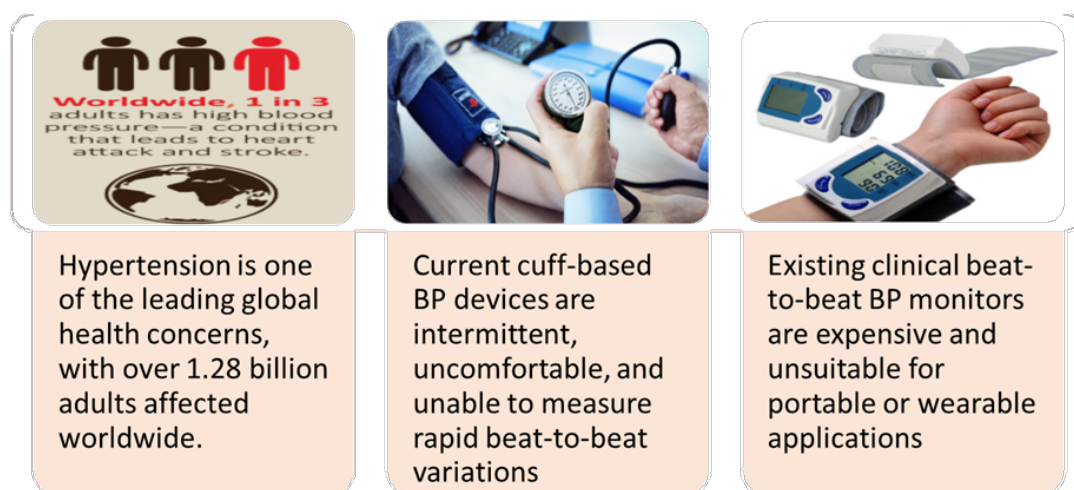


Figure 1: Prevalence of hypertension among adults (39-70) from WHO data

The World Health Organization released its second Global Hypertension Report in September 2025, revealing updated worldwide data on high blood pressure. About 1.4 billion adults aged 30–79 had hypertension in 2024, affecting roughly 33% of people in that age group globally

1.2 Blood Pressure Measurement

Blood pressure can be measured using invasive as well as non-invasive techniques. Invasive blood pressure measurement involves the direct analysis of arterial pressure by manually inserting a catheter equipped with a pressure sensor into the heart or a blood vessel. This method provides continuous and highly accurate measurements and is therefore internationally recognized as the gold standard for blood pressure monitoring.

Despite its accuracy, invasive measurement is primarily used in intensive care units (ICUs) and during surgical procedures, as it is intrusive in nature and carries several risks, including infection, bleeding, and vascular injury. Consequently, it is suitable only for critically ill patients in hospital settings. In this method, the catheter is inserted directly into an artery, allowing real-time measurement of blood pressure.

Non-invasive blood pressure measurement techniques have been developed to overcome these limitations. The first ambulatory blood pressure monitoring (ABPM) systems were introduced over 50 years ago, with early work reported by Hinman et al. (1962) and Sokolow et al. (1966). Several non-invasive approaches have since been proposed, including the auscultatory method, arterial tonometry, volume clamp method, pulse wave analysis, and estimation of mean blood pressure from the pressure–volume relationship of blood vessels.

More recently, methods based on pulse transit time (PTT) or pulse arrival time (PAT) have gained attention. These techniques rely on the relationship between pulse wave velocity and arterial stiffness, enabling indirect estimation of blood pressure without the need for a cuff.

1.3 NIBP(Non-Invasive Blood Pressure) Standards

Worldwide, several apex professional bodies provide recommendations for the standardization and validation of automatic blood pressure (BP) monitoring devices. The most prominent among these are the Association for the Advancement of Medical Instrumentation (AAMI), the British Hypertension Society (BHS), and the European Society for Hypertension (ESH).

The first protocol developed by the American National Standards Institute (ANSI) / AAMI (SP10) was published in 1987 and subsequently updated in 1993 and 2002. The British Hypertension Society introduced its validation protocol, which was later revised in 1993. The most recent ANSI/AAMI standard was released in 2009 and further revised in 2013 in collaboration with the International Organization for Standardization (ISO), resulting in the current standard ISO 81060-2:2013. For validation of an automated BP monitor against an auscultatory reference sphygmomanometer, both the BHS and ANSI/AAMI/ISO protocols recommend a minimum of 255 measurements, with at least three measurements obtained from a minimum of 85 subjects. The minimum time interval between successive measurements should be one minute. Measurement accuracy is evaluated using parameters such as mean error (ME), standard deviation error (SDE), and mean absolute error (MAE).

According to AAMI recommendations, the automated device should not differ from the mercury sphygmomanometer—considered the gold standard for non-invasive blood pressure (NIBP) measurement—by a mean difference greater than 5 mmHg, nor by a standard deviation exceeding 8 mmHg for both systolic blood pressure (SBP) and diastolic blood pressure (DBP). An automatic NIBP monitoring device typically consists of five functional components:

- a pneumatic compression cuff.
- a cuff inflation and control mechanism, usually a small electrically driven pump.
- a cuff deflation mechanism.
- a pressure transduction unit for cuff pressure measurement.

- a sensor signal processing and display system for direct readout..

The three most commonly used commercial standalone NIBP monitoring techniques are oscillometric, plethysmographic, and cuff less methods. In these approaches, a proportional electrical signal is generated either from the inflated cuff pressure or from blood volume changes in peripheral body regions. This signal is appropriately conditioned and digitally processed to compute SBP, DBP, and mean arterial pressure (MAP). Commercial table-top devices provide a direct digital readout after each measurement cycle. Furthermore, with advances in information and communication technologies, the processed signals can be transmitted wirelessly to remote users, enabling continuous and remote blood pressure monitoring.

Table 1: British Hypertension Society Grading Criteria

Grade	Absolute difference between standard and test device (%)		
	≤ 5	≤ 10	≤ 15
A	60	85	95
B	50	75	90
C	40	65	85
D	Worse than C		

1.4 MIMIC II/III Database

The MIMIC-II/III waveform database contains over three million hours of physiological waveform data distributed across approximately 30,000 records. Each record includes time-aligned physiological waveforms such as respiration, finger photo plethysmography (PPG), arterial blood pressure (ABP), and electrocardiogram (ECG) signals, sampled at a frequency of 125 Hz.

The database comprises both waveform records and corresponding numerical records, which have been matched and temporally aligned with the MIMIC-II/III clinical database. In many cases, each recording consists of two components: a waveform record and a corresponding numeric record. A subset of the database includes fully synchronized waveform and numeric data within a single record directory.

For the present study, data from 250 patients were selected from the MIMIC-II/III database. The patients were categorized into three age groups: 15–30 years, 30–45 years, and 45–70 years. For each patient, 30-minute segments of PPG and ABP signals were extracted for analysis.

Despite its richness, the database has certain limitations. In some records, numerical data or waveform signals are missing, and in others, signals may not be perfectly time-aligned. Additionally, a given record may not contain all signal types for the entire duration of the recording.

1.5 Photoplethysmography (PPG) Signal

Photoelectric plethysmography, commonly known as photo plethysmography (PPG), is an optical technique used to measure blood volume changes in peripheral body tissues such

as the fingertips, toes, earlobes, and forehead. The most widespread clinical application of PPG is pulse oximetry, which is routinely used in intensive care units (ICUs) to measure arterial oxygen saturation.

Over the past two decades, extensive research on PPG has demonstrated its significant potential for assessing various physiological parameters, including cardiac output, blood pressure, respiratory activity, heart rate, and other cardiovascular functions.

PPG sensing is based on two fundamental measurement configurations: transmission mode and reflection mode. In both configurations, a matched light-emitting diode (LED)–photodiode (PD) pair is placed on the skin to illuminate a small tissue volume and detect blood volume variations based on the differential absorption of light between blood-filled and blood-depleted tissue, typically at near-infrared (IR) wavelengths.

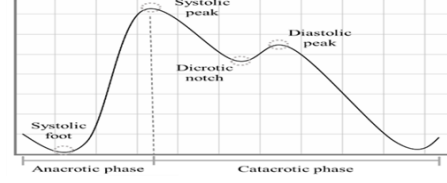


Figure 2: The shape of a typical PPG signal

In transmission-mode PPG, the LED (operating in the wavelength range of 0.8–1 μm) and the photodiode are positioned on opposite sides of the tissue. The photodiode detects light that has passed through bones, soft tissue, and blood capillaries. At this wavelength range, light absorption by tissue water content is minimal. Transmission-mode PPG is feasible only for tissue thicknesses of a few centimetres and is therefore commonly used at sites such as the fingers and earlobes, where sufficient signal-to-noise ratio (SNR) can be achieved.

The PPG waveform consists of two components: a steady (DC) component and a pulsatile (AC) component. The DC component reflects thermoregulatory and autonomic nervous system activity, while the pulsatile component corresponds to arterial blood volume changes associated with the cardiac cycle. The pulsatile waveform comprises two distinct phases: the anacrotic phase, associated with ventricular systole, and the catacrotic phase, associated with ventricular diastole. These phases are characterized by four key fiducial points: the foot (F), systolic peak (SP), dicrotic notch (DN), and diastolic peak (DP).

During ventricular systole, a transient increase in peripheral blood volume leads to greater absorption of incident light by blood, resulting in reduced light intensity at the photodiode. This light absorption phenomenon can be quantitatively described by the Beer–Lambert law:

$$I = I_0 e^{-d\alpha c} \quad (1)$$

where I is the light intensity at a distance d , I_0 is the intensity at source, α is the absorption coefficient of blood, and c is the concentration of the medium. It should be noted that this simplified model neglects additional effects such as scattering and reflection from bones and surrounding tissues. By convention, the PPG waveform is often inverted so that it exhibits a positive correlation with blood volume.

In the catacrotic phase, reduced blood flow results in increased detected light intensity. The dicrotic notch and diastolic peak arise due to a brief increase in peripheral blood flow caused by the reflection of the pressure wave from the closed aortic valve. Similar to the electrocardiogram (ECG), the morphological features of the PPG waveform provide clinically relevant information related to cardiovascular health.

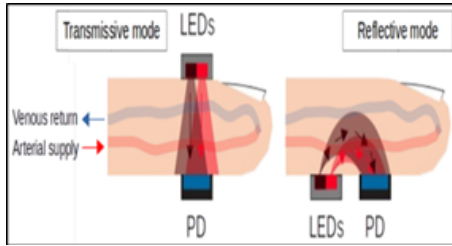


Figure 3: Sensing of PPG signaling: Transmissive Mode & Reflection Mode

flow causes non-hemolyzed erythrocytes to act as microscopic reflectors, redirecting incident light back toward the photodiode.

In reflection-mode PPG, both the LED and photodiode are placed on the same side of the tissue surface, typically using green light ($0.5\text{--}0.6\ \mu\text{m}$), which has gained popularity in recent years. Unlike transmission mode, reflection-mode PPG can be applied to almost any body site. In this configuration, the detected photons follow a banana-shaped light path, penetrating only shallow depths of the tissue. This behaviour occurs because increased blood

This project is structured to provide a comprehensive and systematic approach to continuous, non-invasive blood pressure estimation using photoplethysmography (PPG) signals. It begins with fundamental concepts of blood pressure, its measurement techniques, and standard guidelines, along with an overview of the MIMIC-III Database used for model development. A detailed literature review is then presented to analyze existing methods and identify research gaps.

Subsequently, various signal denoising techniques are explored, including Butterworth, Chebyshev, Elliptic, and Savitzky–Golay filters, to enhance PPG signal quality. Feature extraction methodologies are investigated with a focus on deep learning approaches, particularly the Variational Autoencoder (VAE), along with comparisons to conventional techniques such as Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), and Principal Component Analysis (PCA).

The study further evaluates different computational models for blood pressure estimation, highlighting the effectiveness of Random Forest regression in handling nonlinear physiological data. The proposed methodology, associated challenges, and system design are then discussed, followed by the development of a PPG acquisition circuit for real-time implementation. Finally, the results are analyzed and discussed in detail, demonstrating the performance and feasibility of the proposed system.

CHAPTER 2

2 Literature Review

2.1 Survey of Existing Methods

Over the past several decades, numerous research groups worldwide have invested significant time and effort in the development of non-invasive, cuffless, and continuous blood pressure monitoring techniques. The primary objective of this project is to replace conventional cuff-based blood pressure monitoring devices with a more comfortable and continuous alternative.

Cuff-based blood pressure measurement systems rely on periodic cuff inflation and deflation, which can cause discomfort and inconvenience to patients and often require trained healthcare professionals for proper operation. Additionally, these methods may suffer from measurement inaccuracies and are inherently incapable of providing continuous blood pressure monitoring.

As a result, continuous blood pressure monitoring—which plays a crucial role in the early detection and management of cardiovascular diseases—cannot be effectively achieved using currently available clinical cuff-based devices. This limitation has driven the growing interest in the development of cuffless blood pressure monitoring systems for long-term and real-time cardiovascular health assessment.

Table 2: Some selected previous research papers on blood pressure measurement

Title (Year)	Dataset Used	Feature Extraction and Selection	ML Tools	Performance ¹	
				SBP	DBP
Machine-Learning Models For Predicting Blood Pressure And Cardiac Function Using Wearable Sensor Data (2025)	Cross-sectional dataset (N=180, completeness=0.86); mean SBP=128.4 mmHg, DBP=79.6 mmHg, RMSSD=32.5 ms.	Window-based timing and morphology features (PAT, PPG upstroke time, pulse width, activity intensity) and HRV (RMSSD) derived from PPG morphology. Feature selection based on physiological plausibility, correlation, and regression significance; SHAP ranks features in XGBoost.	Multiple Linear Regression, Random Forest, and XGBoost for SBP, DBP, and RMSSD using subject-wise 5-fold cross-validation with a Python ML pipeline for training and hyperparameter tuning	MAE=5.0, RMSE=7.3, $R^2 = 0.70$	MAE=3.7, RMSE=5.3, $R^2 = 0.57$
Beat-to-Beat Oscillometric Blood Pressure Estimation: A Bayesian Approach With System Identification (2025)	Real data: 135 OMWs from 9 adults (University of Ottawa dataset) with Omron cuff BP reference. Simulated data: 500 OMWs with varied cardio-respiratory parameters for evaluation.	Model-based OMW/OMWE parameter estimation to derive SBP and DBP. All physiological parameters jointly estimated.	Uses Newton-Raphson optimization, EKF, and PF for nonlinear state-space estimation; no machine learning models used.	ME=-1.26, SD=5.95	ME=2.03, SD=4.16

¹All performance values are measured in mmHg.

Table 3: Some selected previous research papers on blood pressure measurement

Title (Year)	Dataset Used	Feature Extraction and Selection	ML Tools	Performance ²	
				SBP	DBP
A Wireless Noninvasive Blood Pressure Measurement System Using MAX30102 and RandomForest Regressor for Photoplethysmography Signals (2024)	WiFi streaming to MySQL (13 samples), 114 BP measurements from 6 subjects with 2856 feature rows.	Four PPG amplitudes—foot, systolic peak, diastolic peak and all four features used.	Multi-output Random Forest regression for SBP/DBP; grid-search tuned (60 trees, max_depth=15).	MAE=4.38, $R^2 = 0.37$	MAE=4.49, $R^2 = 0.46$
Cuff less Blood Pressure Estimation Model Based on Prior Information of Physiological Data and LSTM (2024)	UCI-BP (1200 samples, ICU patients); external test: ~1000 samples from MIMIC-III.	LSTM stack learns temporal features from time- and frequency-domain sequences. Features fused before final regression.	LSTM framework (PFE + 6-layer residual LSTM backbone) with MLP for BP prediction; pretrained using MSE + softDTW loss.	MAE=1.02, SD=1.49	MAE=3.16, SD=3.74
Cuffless BP Estimation Model Using Clustering Technique (2024)	VitalDB (1,051,462 cycles from 1675 subjects); external validation: MIMIC-III (513,623 beats, 236 recordings).	29 handcrafted PWA features from ECG, PPG, APG, and stiffness index. 28 of 41 features retained based on BP correlation.	K-means clustering (K=6) with SHAP; BP estimation using MLR, Extra-Trees, and ANN + BiLSTM RNN.	MAD=3.17, SD=6.43	MAD=2.82, SD=3.65
SMART-BP: SEMResNet and Auto Regressor based on a two-stage framework for non-invasive BP measurement (2024)	Mindray (683 patients) and MIMIC (20k PPG-BP records); invasive BP used as reference.	75 PPG-based morphological, time, frequency, and nonlinear features. PMFL algorithm with ranking, thresholding, and recursive elimination.	SEM-ResNet for BP classification; H2O AutoML stacked ensemble for regression with transfer learning.	MAE=2.52, SD=3.26, ME=0.01	MAE=1.47, SD=2.22, ME=-0.01
Cuffless Blood Pressure Estimation Using Dual Physiological Signal and Its Morphological Features (2023)	UCI cuffless BP (MIMIC-II) with 1612 cleaned samples from ~1000 subjects, stratified into three BP ranges.	Nine physiological features from PPG/derivative and PAT. Fixed nine features chosen based on physiological relevance; interaction terms added.	Gaussian SVM ($\approx 87.8\%$ accuracy) for three BP classes. Multiple Linear Regression (MLR) with three SBP subclass models.	MAE=4.46, SD=3.20, RMSE=5.27	MAE=4.20, SD=3.28, RMSE=4.70
Cuffless Blood Pressure Measurement Using Smartwatches: A Large Scale Validation (2023)	CAS-BP dataset—29,568 recordings from 3,077 adults; dual-observer auscultation BP reference (ANSI/AAMI/ISO 81060-2:2019).	Demographics with ECG/PPG-derived physiological features. Deep learning models use normalized 5-s signal segments.	PTT formulas; traditional ML (Ridge, SVM, AdaBoost, RF); deep learning models (VGG16, ResNet50, BiLSTM, ResLSTM).	MAE=7.38, SD=9.57, ME=2.31	MAE=5.13, SD=6.43, ME=1.33

²All performance values are measured in mmHg.

Table 4: Some selected previous research papers on blood pressure measurement

Title (Year)	Dataset Used	Feature Extraction and Selection	ML Tools	Performance ³	
				SBP	DBP
Causal Inference Based Cuffless Blood Pressure Estimation: A Pilot Study (2023)	Custom single-center dataset—62 adults with 10-minute recordings (mean $SBP114 \pm 13.4 \text{ mmHg}$, $DBP67 \pm 10.6 \text{ mmHg}$).	222 handcrafted ECG/PPG features (PTT, durations, widths, amplitudes, areas, ratios). FCI + PCMCI causal analysis used to select a small causal feature subset.	Multiple Linear Regression and XGBoost using causal features; PTT-only models used as baseline. Leave-one-subject-out cross-validation used for evaluation.	MAD=5.10, SDE=6.44, ME=0.43	MAD=2.85, SDE=3.89, ME=0.57
Causal Inference Based Cuffless Blood Pressure Estimation: A Pilot Study (2023)	Custom single-center dataset—62 adults with 10-minute recordings (mean $SBP114 \pm 13.4 \text{ mmHg}$, $DBP67 \pm 10.6 \text{ mmHg}$).	222 handcrafted ECG/PPG features (PTT, durations, widths, amplitudes, areas, ratios). FCI + PCMCI causal analysis used to select a small causal feature subset.	Multiple Linear Regression and XGBoost using causal features; PTT-only models used as baseline. Leave-one-subject-out cross-validation used for evaluation.	MAD=5.10, SDE=6.44, ME=0.43	MAD=2.85, SDE=3.89, ME=0.57
BePCon: A Photoplethysmography Based Quality Aware Continuous Beat-to-Beat Blood Pressure Measurement Technique Using Deep Learning (2022)	150 MIMIC-II/III records; 50:50 train-test split with 10-fold cross-validation.	50 features per beat (time, DWT, statistical, SDAE); 200 features with 4-beat context. RFE with Random Forest reduced to 20 features per beat.	Temporal Convolutional Network (TCN) with dilated residual blocks for SBP and DBP regression.	MAE=2.38, SD=3.24	MAE=1.23, SD=1.73
Estimation of Beat-by-Beat Blood Pressure and Heart Rate From ECG and PPG Using a Fine-Tuned Deep CNN Model (2022)	UCI cuffless BP dataset (MIMIC-II) with 1551 patients after filtering from 12,000 records.	End-to-end CNN-LSTM learns features directly from raw ECG and PPG signals.	Two-scale CNN + LSTM (modified LRCN) for SBP, DBP, and HR regression.	MAE=3.46, SD=5.90	MAE=2.01, SD=3.65
PPG Based BP estimation combining Filter-Wrapper collaborated feature selection with LASO-LSTM Model (2021)	UCI-ML-BP (10,162 subjects) and MPC-FAHUSTC (200 patients).	54 handcrafted features (morphology, gradient, time, amplitude, HRV). NMI-MCMR filter + RF wrapper (IFS) → 15 optimal features.	Benchmark—MLR, SVM, RF, DTR, KNN, ANN, AdaBoost; Proposed—LASSO-LSTM.	MAE=4.95, SD=4.15	MAE=3.15, SD=2.47
Estimating Blood Pressure from the Photoplethysmogram Signal and Demographic Features Using Machine Learning Techniques (2020)	219 subjects; cuff BP reference; 85% train/validation, 15% test.	107 features (75 time, 16 frequency, 10 statistical, 6 demographic). ReliefF to reduce complexity and overfitting.	Gaussian Process Regression (GPR) with ReliefF-selected features for SBP/DBP regression.	MAE=3.02, RMSE=6.74, MSE=45.49, R=0.95	MAE=1.74, RMSE=3.59, MSE=12.89, R=0.96

Table 5: Some selected previous research papers on blood pressure measurement

Title (Year)	Dataset Used	Feature Extraction and Selection	ML Tools	Performance ⁴	
				SBP	DBP
End-To-End Deep Learning Architecture for Continuous Blood Pressure Estimation Using Attention Mechanism (2020)	Proprietary—15 healthy adults, 30 min recordings (Finometer Pro & BIOPAC); MIMIC not used for training.	CNN (10 conv layers) learns features from raw signals.Implicit via CNN + Bi-GRU + attention over time steps.	VGG-like 1D CNN → Bi-GRU → self-attention → perceptron for SBP/DBP prediction.	MAE=4.06, STD=4.04, $R^2 = 0.52$	MAE=3.33, STD=3.42, $R^2 = 0.49$
Beat-to-Beat Continuous Blood Pressure Estimation Using Bidirectional Long Short-Term Memory Network (2020)	Custom dataset; mean BP 111.2/67.7 mmHg.	7 handcrafted features (4 intervals, 3 amplitudes).Signal combinations compared.	Bidirectional LSTM regression for continuous SBP/DBP.	MAE=5.82, RMSE=6.82	MAE=5.24, RMSE=6.06

2.2 Research Gap

Existing studies primarily focus on machine learning-based blood pressure estimation using ECG and PPG datasets collected under controlled experimental conditions. While these approaches often achieve high estimation accuracy, significant gaps remain in terms of practical hardware implementation, real-time signal processing, and robust performance in the presence of noise and motion artifacts.

These limitations highlight the need for developing a fully embedded, real-time, beat-to-beat blood pressure estimation system that can operate reliably in real-world, ambulatory environments.

³All performance values are measured in mmHg.

⁴All performance values are measured in mmHg.

CHAPTER 3

3 Denoising Tools

Denoising tools are used in PPG signals because the recorded signal is easily contaminated by motion artifacts, baseline drift, power-line interference, ambient light, and sensor noise. These unwanted components distort the pulse waveform and reduce the accuracy of important parameters such as heart rate, SpO_2 , and blood pressure. Denoising improves the signal-to-noise ratio, preserves pulse morphology, and ensures reliable beat detection and physiological measurement, especially in real-time and wearable applications.

3.1 Butterworth Filter

The Butterworth filter is a type of signal processing filter designed to have a frequency response that is as flat as possible in the passband. It is also referred to as a maximally flat magnitude filter.

Like all filters, the typical prototype is the low-pass filter, which can be modified into a high-pass filter, or placed in series with others to form band-pass and band-stop filters, and higher order versions of these.

The gain $G(\omega)$ of an n^{th} order butterworth low pass filter is given in terms of the transfer function $H(s)$ as

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}} \quad (2)$$

where n is the order of the filter, ω_c is the cut-off frequency (approximately the -3dB frequency), and $G_o (= 1)$ is the D.C gain (gain at zero frequency).

We wish to determine the transfer function $H(s)$ where $s = \sigma + j\omega$. Because $|H(j\omega)|^2 = H(s) \cdot H(\bar{s})$ and as a general property of Laplace Transform at $s = j\omega$, $H(-j\omega) = H(j\omega)$. If we choose $H(s)$ such that:

$$H(s) \cdot H(-s) = \frac{(G_o)^{2n}}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}} \quad (3)$$

The n poles of this expression occur on a circle of radius ω_c at equally-spaced points, and symmetric around the negative real axis. For stability, the transfer function $H(s)$ is therefore chosen such that it contains only the poles in the negative real half of s plane. The k^{th} pole is specified by,

$$\begin{aligned} \frac{-s_k^2}{\omega_c^2} &= (-1)^{\frac{1}{n}} = e^{\frac{j(2k-1)\pi}{n}} \\ \Rightarrow s_k &= \omega_c e^{\frac{j(2k+n-1)\pi}{2n}}, \quad k = 1, 2, 3, \dots, n \end{aligned}$$

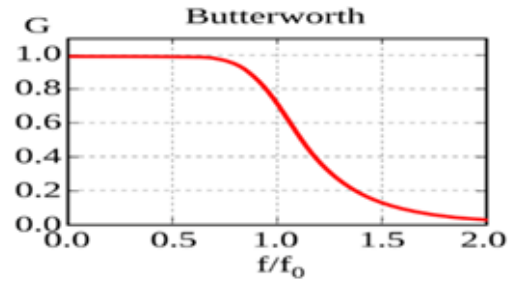


Figure 4: Graphical Representation of Filter Transfer Function

The transfer (or system) function may be written in terms of these poles as

$$H(s) = G_0 \prod_{k=1}^n \frac{\omega_c}{s - s_k} = G_0 \prod_{k=1}^n \frac{\omega_c}{s - \omega_c e^{\frac{j(2k+n-1)\pi}{2n}}} \quad (4)$$

where $\prod$ is the product of a sequence operator. The denominator is a Butterworth polynomial in s .

3.2 Chebyshev Filter

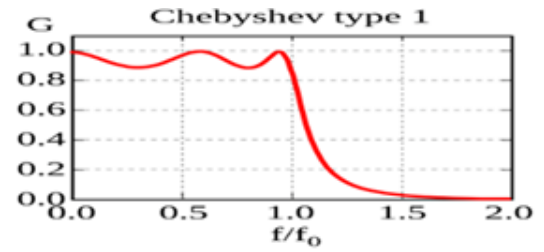
A Chebyshev filter is a class of analog filter whose magnitude response is based on Chebyshev polynomials. Compared to the Butterworth filter, it provides a sharper roll-off for the same filter order, but at the cost of ripples in either the passband or the stopband. Chebyshev filters are of two types:

- Type-I (Chebyshev-I)
- Type-II (Chebyshev-II or Inverse Chebyshev)

3.2.1 Chebyshev Type-I Filter

A Chebyshev Type-I filter has:

- Equiripple passband
- Monotonic (no ripple) stopband
- Faster roll-off than Butterworth filter



Magnitude Response:

The squared magnitude response is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \varepsilon^2 T_n^2\left(\frac{\omega}{\omega_0}\right)} \quad (5)$$

Figure 5: Graphical Representation of Filter Transfer Function

where ε is ripple factor (controls passband ripple), ω_c is cutoff frequency, n is the order of the filter, $T_n(x)$ is chebyshev polynomial of degree n

Chebyshev Polynomial:

$$T_n(x) = \begin{cases} \cos(n \cos^{-1} x) & \text{if } |x| \leq 1 \\ \cosh(n \cosh^{-1} x) & \text{if } |x| > 1 \end{cases} \quad (6)$$

Transfer Function (Type I):

The transfer function is written as:

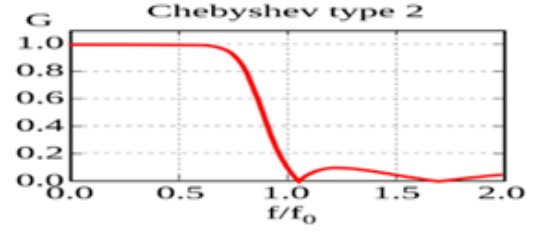
$$H(s) = \frac{G_o}{\prod_{k=1}^n (s - s_k)} \quad (7)$$

where s_k are the left half plane poles obtained from the chebyshev response

3.2.2 Chebyshev Type-II Filter

A Chebyshev Type-II filter has:

- Monotonic passband
- Equiripple stopband
- Transmission zeros in the stopband



Magnitude Response:

The squared magnitude response is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \frac{1}{\varepsilon^2 T_n^2\left(\frac{\omega_0}{\omega}\right)}} \quad (8)$$

where ε is stopband ripple factor, $T_n(\cdot)$ is chebyshev polynomial, ω_c is cutoff frequency

Figure 6: Graphical Representation of Filter Transfer Function

Transfer Function(Type II):

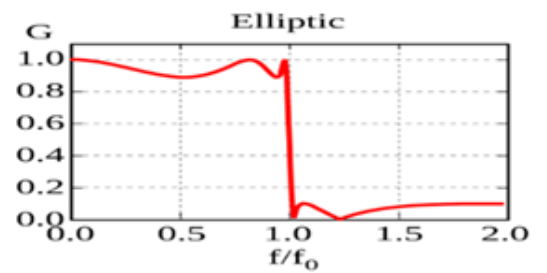
The transfer function is written as:

$$H(s) = G_o \frac{\prod_{k=1}^n (s - z_k)}{\prod_{k=1}^n (s - p_k)} \quad (9)$$

where p_k poles in the left half plane, z_k is finite transmission zeros (in stopband)

3.3 Elliptic Filter

An Elliptic filter, also known as a Cauer filter, is a class of analog filter that provides the sharpest transition between passband and stopband for a given filter order. This is achieved by allowing ripples in both the passband and the stopband.



Magnitude Response:

The squared magnitude response of an elliptic low pass filter is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \varepsilon^2 R_n^2\left(\frac{\omega}{\omega_c}, k\right)} \quad (10)$$

where ε is passband ripple factor, ω_c is cutoff frequency, n is order of the filter, $R_n(\cdot, k)$ is Jacobian elliptic rational function, k is elliptic modulus(controls stopband ripple)

Figure 7: Graphical Representation of Filter Transfer Function

Transfer Function of Elliptic Filter:

The transfer function is written as:

$$H(s) = G_0 \frac{\prod_{k=1}^n (s - z_k)}{\prod_{k=1}^n (s - p_k)} \quad (11)$$

where p_k is poles (lie in the left half of the s-plane for stability), z_k is the finite transmission zeros (located on the imaginary axis for low pass filters), G_o is gain constant

3.4 Savitzky-Golay Filter

The Savitzky-Golay (SG) filter doesn't have a single, simple transfer function like standard filters because it's a Finite Impulse Response (FIR) filter based on local least-squares polynomial fitting within a moving window, not a fixed difference equation. Its "transfer function" is implicitly defined by the specific polynomial coefficients (weights) derived for a given window size and polynomial order, which are essentially a weighted average but preserve high frequencies better than simple moving averages. It like a bank of FIR filters, where each center point's output is found by fitting a polynomial (e.g., quadratic, cubic) to its neighbors and evaluating the polynomial at the center. It's conceptual working mechanism

- **Windowing:** A fixed-size "window" slides over your data.
- **Polynomial Fitting:** Within each window, a polynomial of degree n is fitted to the data points using least-squares
- **Evaluation:** The value/derivative at the center of the window is determined by evaluating the fitted polynomial at that point, not by averaging the window's points.
- **Coefficients:** The weights/coefficients we see are the result of this process, effectively creating a weighted sum for the center point, but with non-uniform, intelligently chosen weights.

Filter Operation:

For a window of length $(2M + 1)$, the output is

$$y[n] = \sum_{k=-M}^M h[k] x[n - k] \quad (12)$$

where $h[k]$ are the Savitzky-Golay filter coefficients and M is half window length.

Transfer Function of Savitzky-Golay Filter:

Since the Savitzky-Golay filter is an FIR filter, its transfer function in the z domain is:

$$H(z) = \sum_{k=-M}^M h[k] z^{-k} = z^{-M} \sum_{k=0}^{2M} h[k - M] z^{-k} \quad (13)$$

CHAPTER 4

4 Feature Extraction Techniques

In feature extraction we transform the original features into a new smaller set of features. The new features are combinations or projections of the originals so they are often less directly interpretable.

4.1 Autoencoder

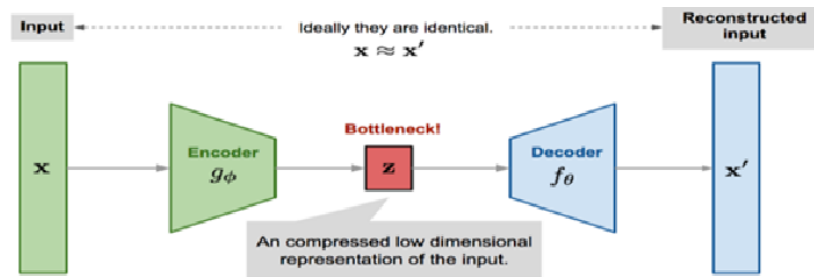


Figure 8: Block Diagram of An Autoencoder

An autoencoder is an unsupervised neural network used to learn an efficient representation of data by encoding the input into a lower-dimensional space and then reconstructing it at the output. The goal of an autoencoder is to minimize the reconstruction error between the input and the output.

Architecture of Autoencoder:

An autoencoder consists of three main parts:

- **Encoder:** The encoder maps the input vector x to a latent representation z . Therefore it performs dimensionality reduction.
- **Latent Space (Bottleneck Layer):** This is a compressed representation of the input containing the most important features.
- **Decoder:** The decoder reconstructs the original input from the latent vector

Working Principle

- The input and output are identical.
- The network is trained to minimize the reconstruction loss, (commonly Mean Squared Error)

$$L = \|x - \hat{x}\|^2 \quad (14)$$

- Training is done using backpropagation and gradient descent.

Types of Autoencoders

- **Undercomplete Autoencoder:** latent space smaller than input
- **Denoising Autoencoder:** removes noise from corrupted input
- **Sparse Autoencoder:** enforces sparsity constraint
- **Variational Autoencoder (VAE):** probabilistic and generative
- **Convolutional Autoencoder:** used for images and signals

Applications

- Dimensionality reduction
- Noise removal in signals (ECG, PPG)
- Feature extraction
- Anomaly detection
- Image and signal compression

Advantages

- Learns features automatically
- Works in an unsupervised manner
- Effective for non-linear data

Limitations

- Can learn identity mapping if not properly constrained
- Requires large datasets
- Latent features may be hard to interpret

4.1.1 Variational Autoencoder

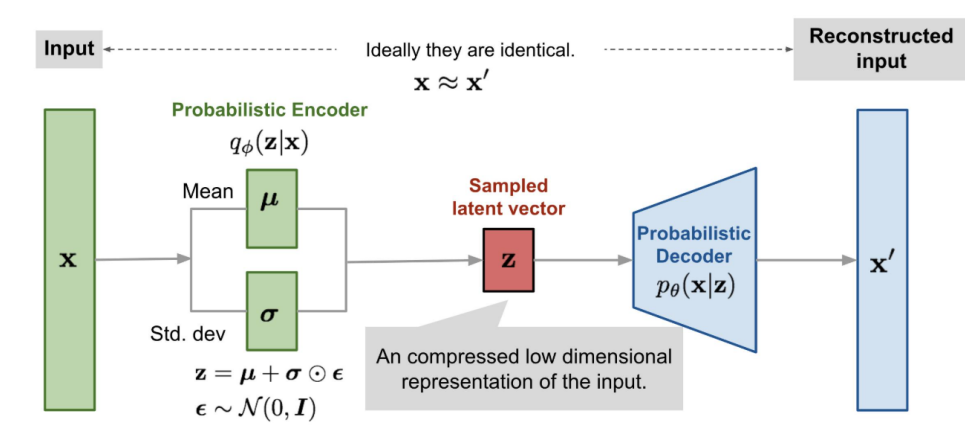


Figure 9: Block Diagram of A Variational Autoencoder

A Variational Autoencoder is a probabilistic generative model (or neural network) that learns a compressed latent representation of input data by mapping it to a distribution rather than a fixed vector. It consists of an encoder that outputs the mean and variance of a latent Gaussian distribution, and a decoder that reconstructs the input from sampled latent variables. The model is trained using a loss function consisting of reconstruction loss and KL divergence, which ensures both accurate reconstruction and a structured latent space. In this project, the VAE is used to extract compact latent features from PPG signal segments, which are then used as inputs to a regression model for estimating systolic and diastolic blood pressure.

4.1.2 Reparameterization Trick in Variational Autoencoders

In a Variational Autoencoder (VAE), the encoder network maps an input x to the parameters of a latent Gaussian distribution, namely the mean μ and variance σ^2 . The latent variable z is then sampled as:

$$z \sim \mathcal{N}(\mu, \sigma^2) \quad (15)$$

However, this stochastic sampling operation introduces a challenge during training. Neural networks are optimized using gradient-based methods such as backpropagation, which require all operations in the computational graph to be differentiable. Direct sampling from a distribution is inherently non-differentiable with respect to the parameters μ and σ , as it introduces randomness that breaks the gradient flow.

To overcome this limitation, the *reparameterization trick* is employed. The key idea is to express the random variable z as a deterministic transformation of the parameters μ , σ , and an auxiliary noise variable ϵ drawn from a standard normal distribution:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1) \quad (16)$$

This reformulation separates the stochasticity from the learnable parameters. The randomness is now entirely captured by ϵ , while μ and σ remain deterministic and differentiable. As a result, the sampling process can be incorporated into the computational graph.

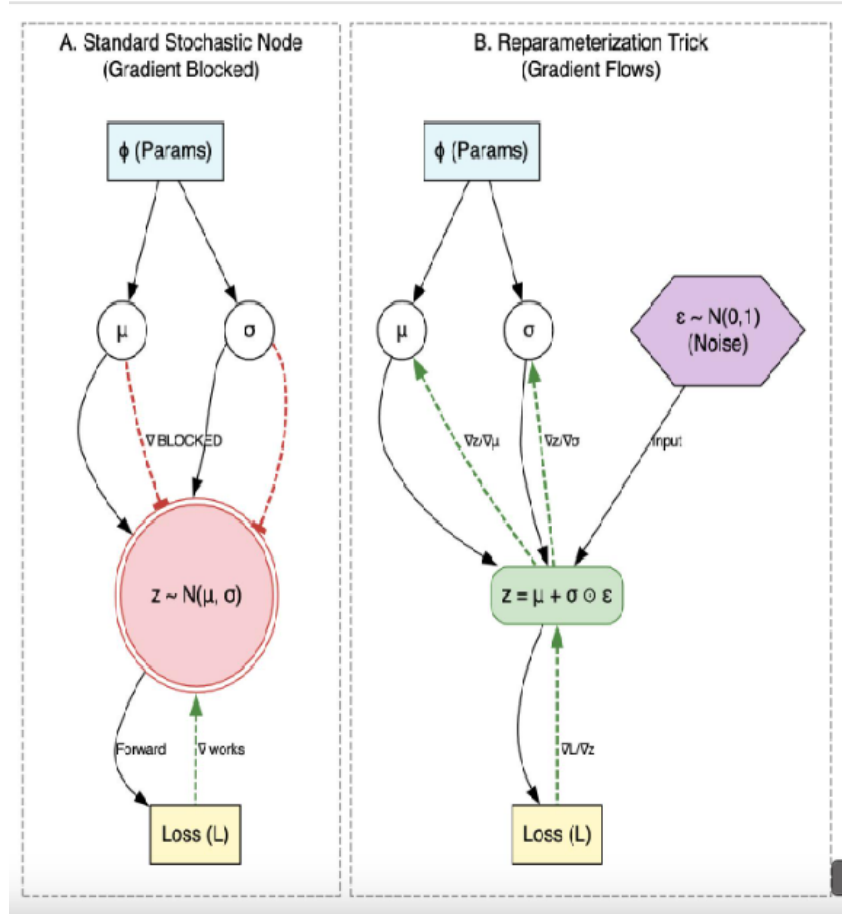


Figure 10: The reparameterization trick.

The transformation involves only elementary operations such as addition and multiplication, which are differentiable. Therefore, gradients can propagate through z with respect to μ and σ , enabling efficient end-to-end training of the VAE using backpropagation. Specifically, the gradients are given by:

$$\frac{\partial z}{\partial \mu} = 1, \quad \frac{\partial z}{\partial \sigma} = \epsilon \quad (17)$$

Thus, the reparameterization trick plays a crucial role in making the variational autoencoder framework trainable by allowing gradient-based optimization despite the presence of stochastic latent variables.

4.1.3 Variational Autoencoder (VAE) Derivation

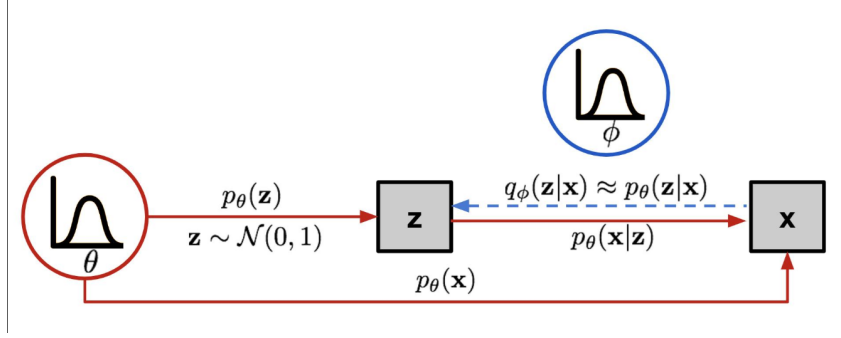


Figure 11: The graphical model involved in Variational Autoencoder. Solid lines denote the generative distribution $p_\theta(\cdot)$ and dashed lines denote the distribution $q_\phi(z|x)$ to approximate the intractable posterior $p_\theta(z|x)$

Aim of Variational Autoencoder

The objective of a Variational Autoencoder (VAE) is to model the data distribution $p(x)$ using a latent variable z . The goal is to maximize the log-likelihood of the observed data:

$$\log p(x)$$

However, this involves computing:

$$p(x) = \int p(x|z)p(z) dz$$

which is intractable due to the high-dimensional integral over z .

Step 1: Introduction of Approximate Posterior

To overcome this issue, an approximate posterior distribution $q(z|x)$ is introduced to approximate the true posterior $p(z|x)$.

Step 2: Reformulation of Log-Likelihood

The log-likelihood can be rewritten as:

$$\log p(x) = \log \int q(z|x) \frac{p(x, z)}{q(z|x)} dz$$

Step 3: Application of Jensen's Inequality

Using Jensen's inequality:

$$\log \mathbb{E}[f(z)] \geq \mathbb{E}[\log f(z)]$$

we obtain:

$$\log p(x) \geq \mathbb{E}_{q(z|x)} \left[\log \frac{p(x, z)}{q(z|x)} \right]$$

Step 4: Evidence Lower Bound (ELBO)

The Evidence Lower Bound (ELBO) is defined as:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q(z|x)} [\log p(x, z)] - \mathbb{E}_{q(z|x)} [\log q(z|x)]$$

Step 5: Expansion of Joint Distribution

Using the identity:

$$p(x, z) = p(x|z)p(z)$$

the ELBO becomes:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q(z|x)} [\log p(x|z)] + \mathbb{E}_{q(z|x)} [\log p(z)] - \mathbb{E}_{q(z|x)} [\log q(z|x)]$$

Rearranging terms:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q(z|x)} [\log p(x|z)] - KL(q(z|x) \parallel p(z))$$

Step 6: KL Divergence for Gaussian Distribution

Assuming:

$$q(z|x) = \mathcal{N}(\mu, \sigma^2), \quad p(z) = \mathcal{N}(0, 1)$$

the KL divergence simplifies to:

$$KL(q(z|x) \parallel p(z)) = \frac{1}{2} \sum (\mu^2 + \sigma^2 - \log \sigma^2 - 1)$$

Step 7: Reparameterization Trick

To enable gradient-based optimization, the latent variable z is reparameterized as:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

This transformation allows the stochastic sampling process to be expressed as a deterministic function, enabling backpropagation.

Hence our loss function for this network will consist of two terms, one which penalizes reconstruction error (which can be thought of maximizing the reconstruction likelihood as discussed earlier) and a second term which encourages our learned distribution $q(z|x)$ to be similar to the true prior distribution $p(z)$, which we'll assume follows a unit Gaussian distribution, for each dimension j of the latent space.

$$\boxed{\mathcal{L} = \mathcal{L}(x, \hat{x}) + KL(q(z|x) \parallel p(z))} \tag{18}$$

4.2 Other Techniques

4.2.1 Discrete Wavelet Transform(DWT)

The DWT is a signal processing technique that analyzes a signal at different frequency ranges with different resolutions, a process known as multiresolution analysis. A wavelet is a short-duration oscillatory waveform with zero average value.

Calculating wavelet coefficient at every possible scale produces a lot of data. If s and τ are chosen to be discrete then the wavelet transform won't generate huge amount of data. If s and τ are based on power of two, so called dyadic scales and position, then analysis become much more efficient and accurate. Therefore, DWT is given by,

$$D[a, b] = \frac{1}{\sqrt{b}} \sum_{m=0}^{\phi-1} f(t) \varphi\left[\frac{t-a}{b}\right] \quad (19)$$

where $a = \tau = k2^{-j}$, $b = s = 2^{-j}$ & $\phi(t) = e^{j\omega t} e^{-\frac{t^2}{2}}$.

It is used in: Compression, Denoising, Discontinuity Detection, Audio Enhancement Effect.

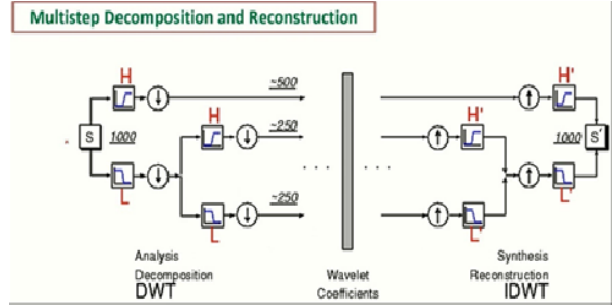


Figure 12: Block Diagram of Decomposition and Reconstruction in DWT

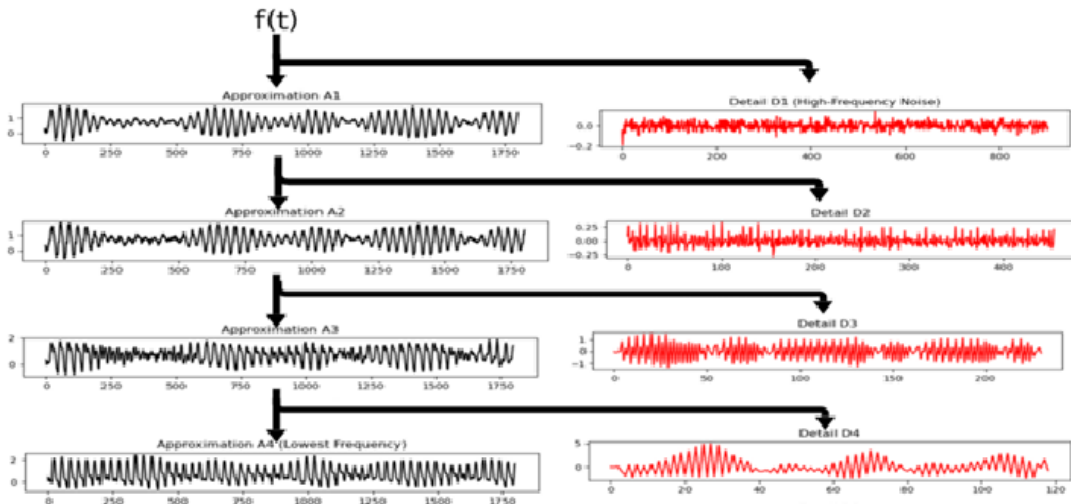


Figure 13: Multilevel Decomposition of A PPG signal

4.2.2 Empirical Mode Decomposition(EMD)

It is a data-driven signal decomposition technique used to analyze nonlinear and non-stationary signals — those whose frequency content changes over time. EMD decomposes a signal $x(t)$ into a set of simpler oscillatory components called Intrinsic Mode Functions (IMFs) and a residual trend. Each IMF represents an oscillation mode with its own instantaneous frequency and amplitude.

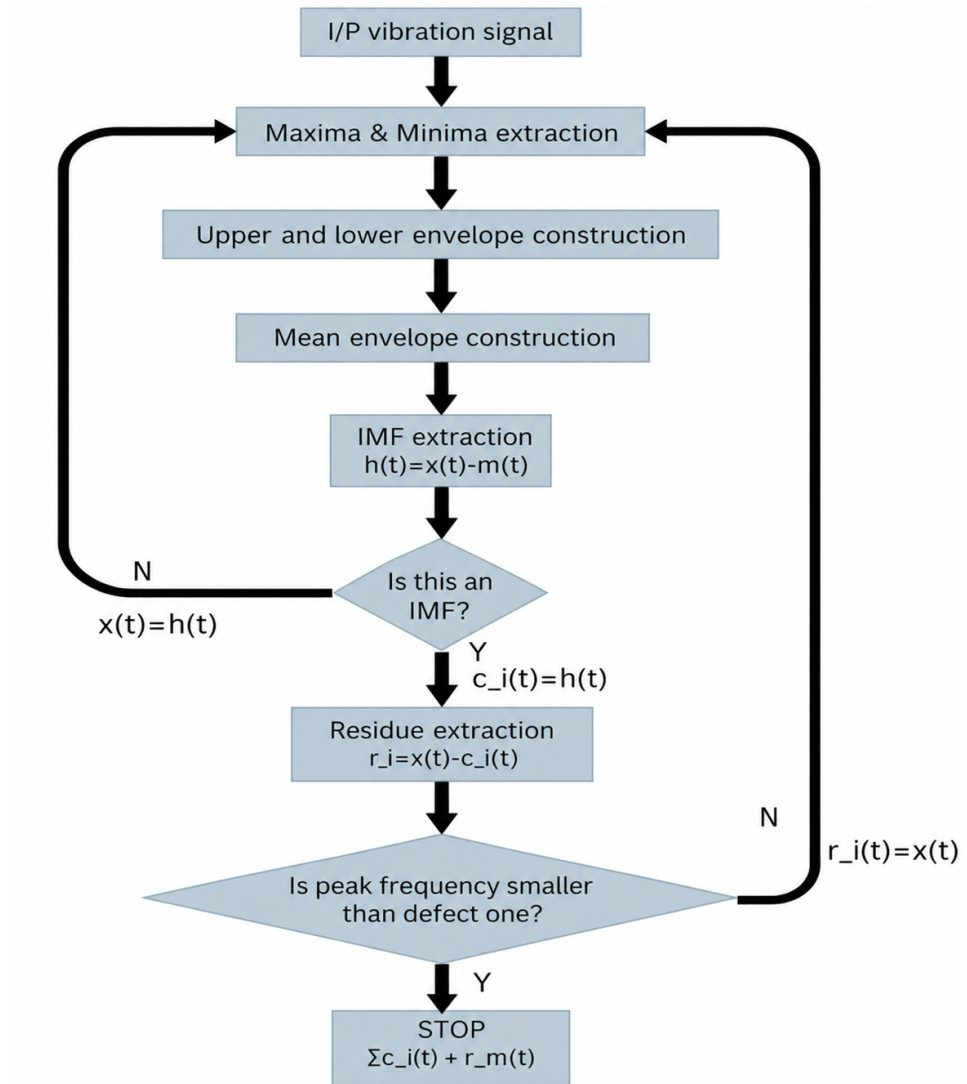


Figure 14: Block Diagram Showing The Algorithm Of EMD

ALGORITHM:

- **Step 1:** Identify all extrema of $x(t)$.
- **Step 2:** Interpolate between minima (respect to maxima), ending up with some envelope $e_{min}(t)$ (respected to $e_{max}(t)$).
- **Step 3:** Compute the mean $m(t) = \frac{e_{min}(t) + e_{max}(t)}{2}$.
- **Step 4:** Extract the details $d(t) = x(t) - m(t)$
- **Step 5:** Iterate 1 to 4 with $d(t)$ until $d(t)$ is zero mean or some stopping criteria, the resulting $d(t)$ is the intrinsic mode function (IMF)
- **Step 6:** Iterate 1 to 5 on residue $m(t) = x(t) - d(t)$ until some stopping criteria like peak frequency is low. The resultant output will be multiple IMF and a residue

It is used in: Component Separation, Noise Reduction

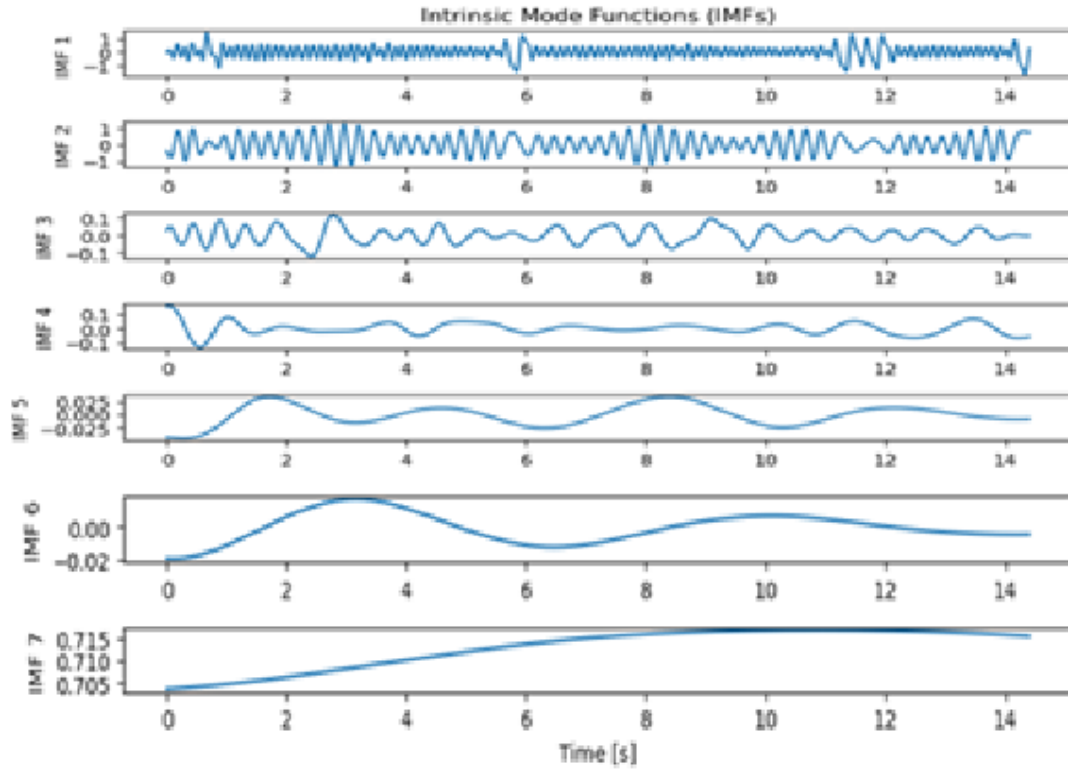


Figure 15: Different IMF Graphical Representation

Table 6: Feature Extraction Table of Different IMFs

IMFs	Mean	Standard Deviation	Energy
IMF 1	-0.0117	0.4224	321.3673
IMF 2	0.0076	0.5587	561.9539
IMF 3	0.0004	0.0434	3.3869
IMF 4	-0.0001	0.0391	2.7521
IMF 5	-0.0010	0.0192	0.6631
IMF 6	-0.0003	0.0090	0.1470
IMF 7	0.7129	0.0044	914.8620

4.2.3 Principal Component Analysis(PCA)

It is a dimensionality reduction technique used in unsupervised learning to transform a data set with very variables it to smaller set that still contains most of the information

ALGORITHM

- **Step 1:** Calculate mean of every feature.
- **Step 2:** Calculate the covariance matrix.
- **Step 3:** Calculate the eigenvalue and eigenvectors.

- **Step 4:** Choose principal components.
- **Step 5:** Project data onto principal component.

ADVANTAGES:

- Reduces dimensionality and computational load.
- Helps denoise the PPG signal by discarding low-variance noise components.
- Improves machine-learning model performance and reduces overfitting.

DISADVANTAGES:

- Captures only linear relationships, missing nonlinear PPG dynamics.
- Highly sensitive to noise, motion artifacts, and improper scaling.
- Requires sufficient data to form stable components.

1) ORIGINAL DATASET AND PPG COMPONENTS

Original shape: (98981, 184)

	SBP	DBP	R	C	PPG_0	PPG_1	PPG_2	\
0	144.5785	73.0708	45.310851	0.028449	0.049658	0.049658	0.049658	
1	142.2829	73.7546	45.268321	0.028441	0.098270	0.098270	0.098270	
2	146.5323	74.7315	44.181763	0.019638	0.090202	0.090202	0.090202	
3	142.6736	72.7289	47.567894	0.019219	0.000000	0.000000	0.000000	
4	141.3548	73.1196	35.442848	0.026788	0.128818	0.127883	0.127615	

	PPG_3	PPG_4	PPG_5	...	PPG_90	PPG_91	PPG_92	PPG_93	\
0	0.049658	0.058131	0.051982	...	0.148739	0.145813	0.143223	0.148979	
1	0.098270	0.098523	0.100191	...	0.128368	0.124864	0.119648	0.115714	
2	0.090164	0.091574	0.094111	...	0.177822	0.172696	0.167699	0.162859	
3	0.000000	0.000000	0.000447	...	0.076328	0.072221	0.068259	0.064435	
4	0.128014	0.130718	0.133548	...	0.140444	0.143845	0.136481	0.135112	

	PPG_94	PPG_95	PPG_96	PPG_97	PPG_98	PPG_99
0	0.139444	0.137426	0.136889	0.134995	0.134897	0.133346
1	0.111644	0.107619	0.103628	0.093628	0.085625	0.081595
2	0.158128	0.153549	0.149128	0.144859	0.140745	0.136794
3	0.068743	0.057175	0.053723	0.058374	0.047112	0.043320
4	0.131290	0.127372	0.123514	0.119678	0.115835	0.111958

[5 rows x 184 columns]

PPG shape: (98981, 100)

(a) Large Data Set of Discrete Time of Arbitrary Signal

2) FINAL OUTPUT AFTER PCA								
FINAL OUTPUT SHAPE: (98981, 12)								
	SBP	DBP	R	C	PC1	PC3	\	
0	144.5785	73.0708	45.310851	0.028449	-3.017828	0.173829	-1.688541	
1	142.2829	73.7546	45.268321	0.028441	-0.397155	3.185632	1.429688	
2	146.5323	74.7315	44.181763	0.019638	0.175765	1.322156	-0.157714	
3	142.6736	72.7289	47.567894	0.019219	-6.381226	0.862584	-1.282485	
4	141.3548	73.1196	35.442848	0.026788	2.285034	4.796949	1.360167	
	PC4	PC5	PC6	PC7	PC8			
0	0.398885	-0.876388	-0.178316	-0.267488	-0.155195			
1	0.698847	-0.434835	-0.168351	-0.366321	-0.216742			
2	0.248124	-0.534277	-0.428873	-0.252424	-0.156811			
3	0.510266	-1.258476	-0.382851	-0.398635	-0.288613			
4	0.278610	0.865281	0.404807	-0.408647	-0.011240			
Saved as train_pca.csv								

(b) Reduced Dimension of Large Data by PCA Algorithm

Figure 16: Data Compression Using Principal Component Analysis

CHAPTER 5

5 Computational Tools

Classifier: A classifier in machine learning is a learned function or model that maps an input feature vector $x \in \mathbb{R}^n$ to one of a finite set of discrete class labels. It is mathematically represented as

$$f : \mathbb{R}^n \rightarrow \{1, 2, \dots, K\},$$

where the function f is learned from data.

Regression: A regression model in machine learning is a learned function or model that maps an input feature vector $x \in \mathbb{R}^n$ to a continuous real-valued output. It is mathematically represented as

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

where the function f is learned from data.

5.1 Random Forest

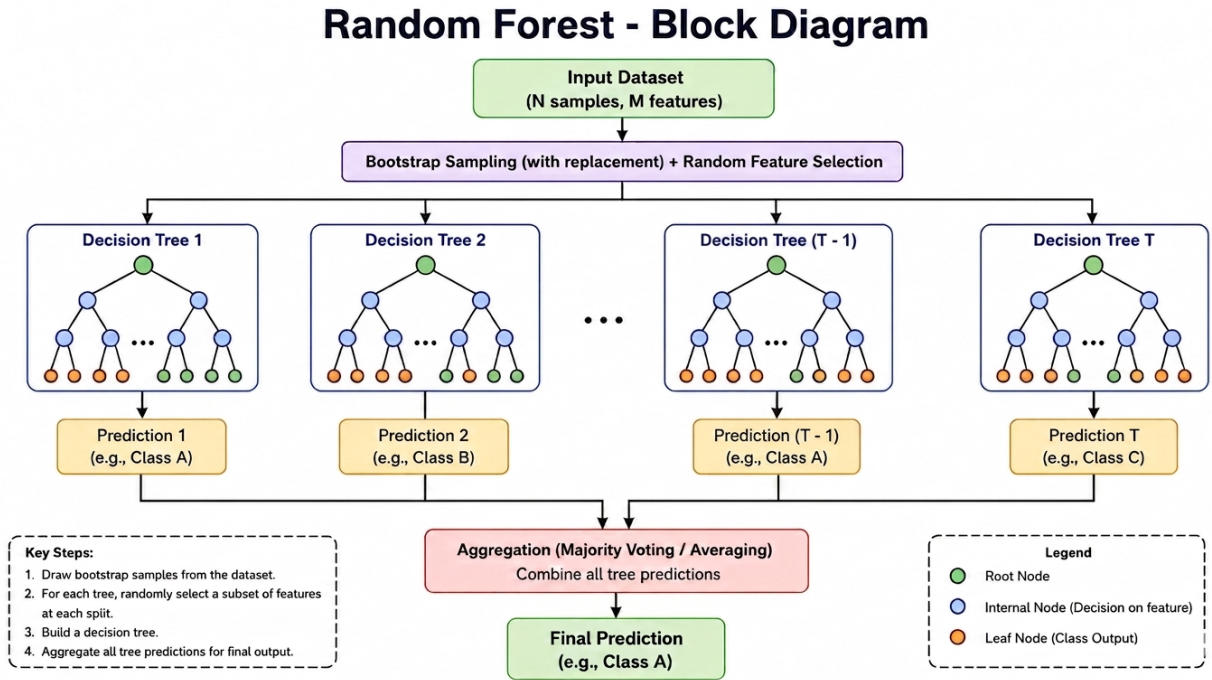


Figure 17: Random Forest

Random Forest is an ensemble learning algorithm that constructs a collection of decision trees and combines their outputs to improve predictive performance and generalization. In blood pressure estimation, the final prediction is obtained by averaging the outputs of individual trees. Mathematically, the Random Forest regression model can be expressed as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(\mathbf{x}) \quad (20)$$

where $T_i(\mathbf{x})$ denotes the prediction of the i^{th} decision tree, $\mathbf{x}$ represents the input feature vector, and N is the total number of trees in the ensemble.

The key principle behind Random Forest is the use of *bootstrap aggregation* (bagging) and random feature selection. During training, multiple subsets of the dataset are generated using random sampling with replacement. Each decision tree is trained on a different subset of the data, and at each split, a random subset of features is considered. This process introduces diversity among the trees, thereby reducing variance and mitigating overfitting.

5.2 Other Tools

5.2.1 Decision Tree

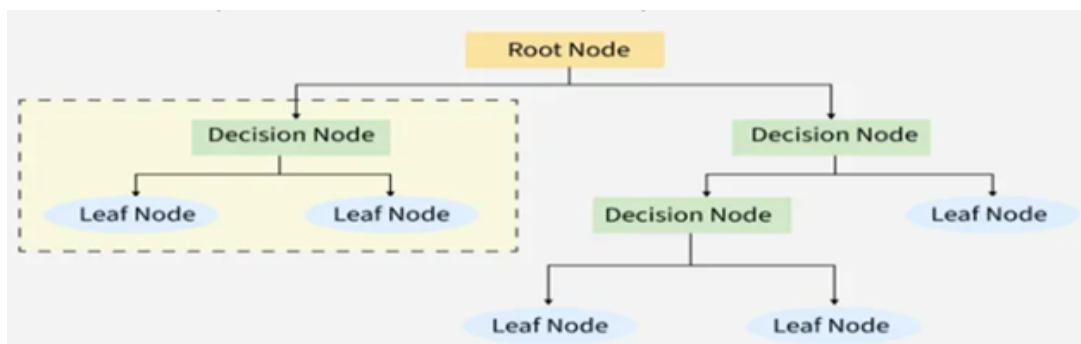


Figure 18: Decision Tree

A Decision Tree is a supervised learning algorithm that splits data into branches based on feature values, forming a tree-like structure of decisions to perform classification or regression.

Advantages:

- Easy to understand and interpret.
- Requires little data pre-processing (no normalization needed).
- Can handle both numerical and categorical data.

Disadvantages:

- Prone to overfitting, especially with deep trees.
- Unstable—small changes in data can change the tree structure.
- Lower accuracy compared to ensemble methods.

5.2.2 Naïve Bayes

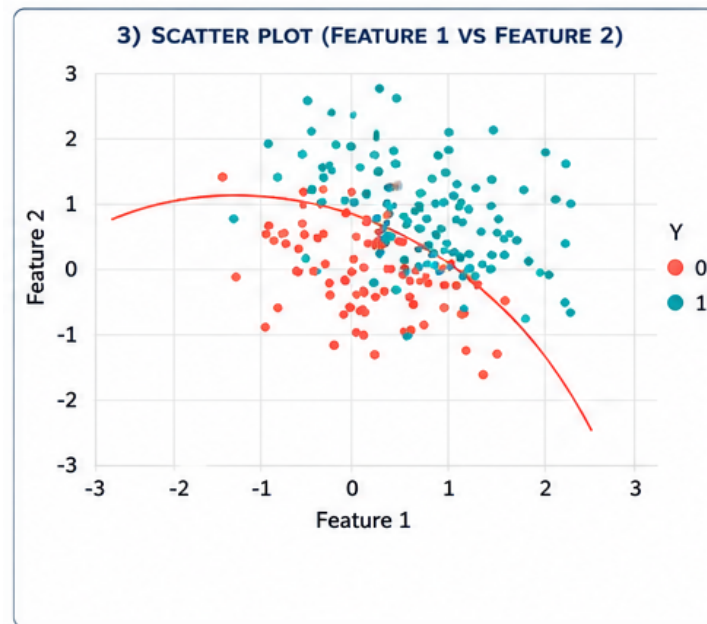


Figure 19: Decision Boundary for Gaussian Naïve Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, assuming that all features are conditionally independent.

Advantages:

- Simple and fast to train and predict.
- Performs well with small datasets.
- Effective for text classification (spam filtering, sentiment analysis).

Disadvantages:

- Assumes feature independence (often unrealistic).
- Poor performance when features are highly correlated.
- Zero probability problem if a class feature is missing.

5.2.3 K-Nearest Neighbors

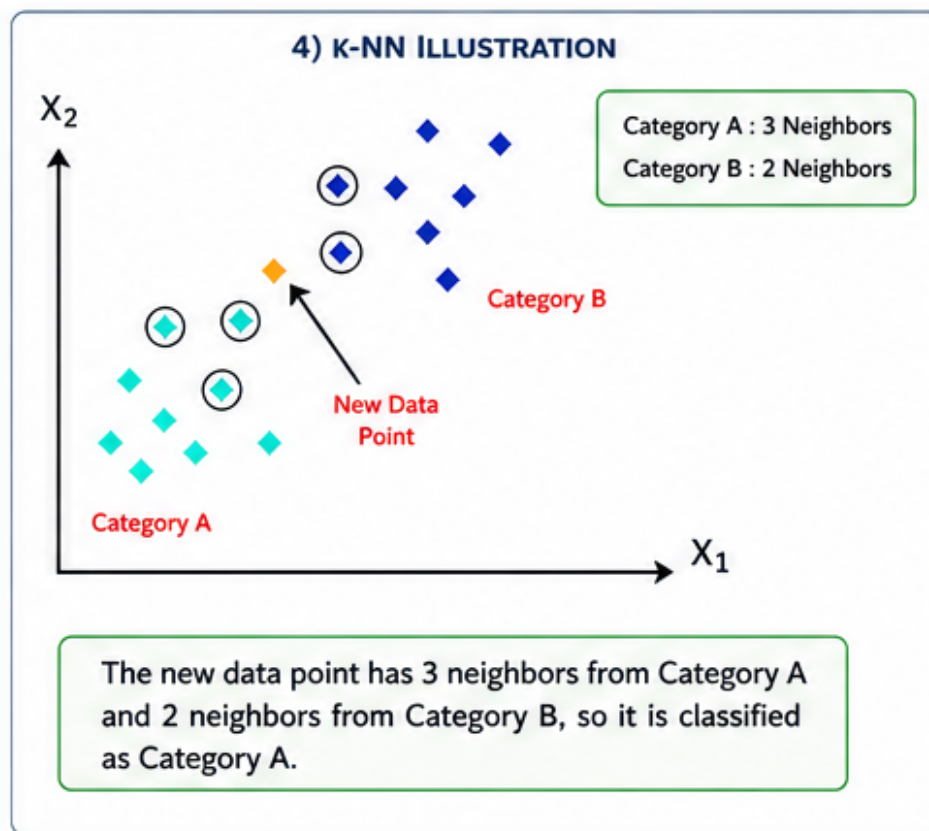


Figure 20: Optimal Value of K in kNN

k-NN is a non-parametric algorithm that classifies data based on the majority class of its k nearest neighbours in feature space.

Advantages:

- Simple and easy to implement.
- No training phase required.
- Works well for small datasets.

Disadvantages:

- Computationally expensive during prediction.
- Sensitive to noise and irrelevant features.
- Requires proper choice of k and distance metric.

5.2.4 Support Vector Machine

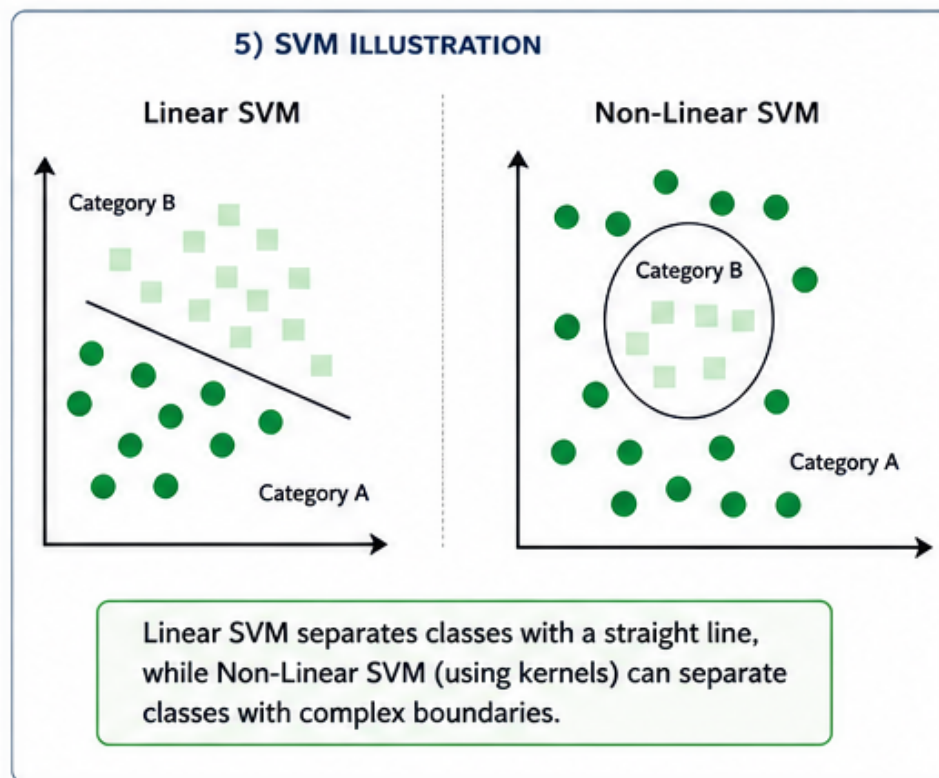


Figure 21: Support Vector Machine Algorithm

SVM is a supervised learning algorithm that finds an optimal hyperplane to separate data points of different classes with maximum margin.

Advantages:

- Effective in high-dimensional spaces.
- Works well with small and medium datasets.
- Kernel trick enables non-linear classification.

Disadvantages:

- High computational cost for large datasets.
- Difficult to choose the right kernel and parameters.
- Less interpretable than tree-based models.

5.2.5 XGBoost

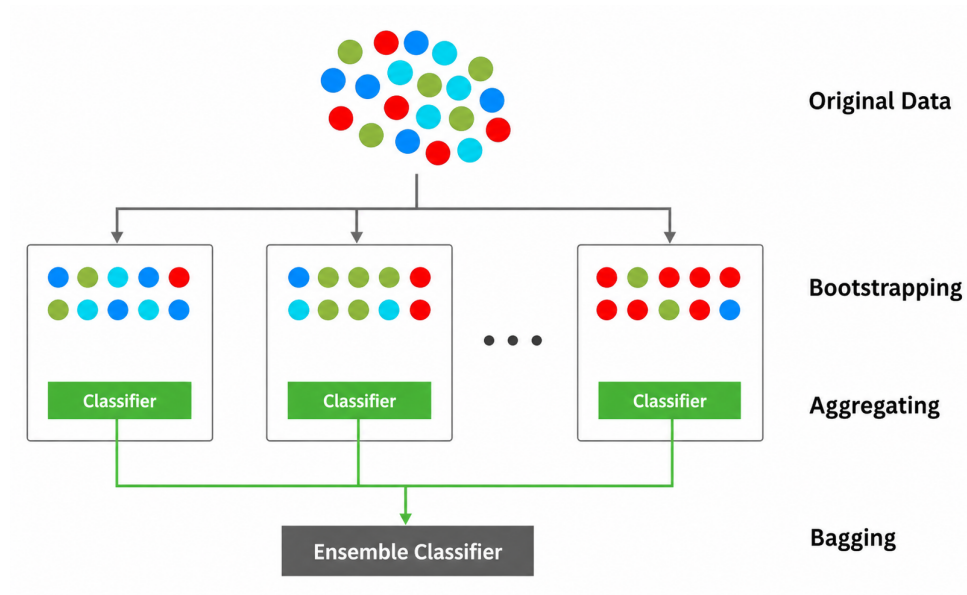


Figure 22: XGBoosting

XGBoost is an optimized gradient boosting algorithm that builds models sequentially, where each new model corrects the errors of the previous ones.

Advantages:

- Very high predictive accuracy.
- Handles missing values efficiently.
- Fast and scalable due to parallel processing.

Disadvantages:

- Complex and difficult to interpret.
- Requires careful hyperparameter tuning.
- Computationally intensive.

CHAPTER 6

6 Proposed Work and Challenges

6.1 Outline of the Proposed Work

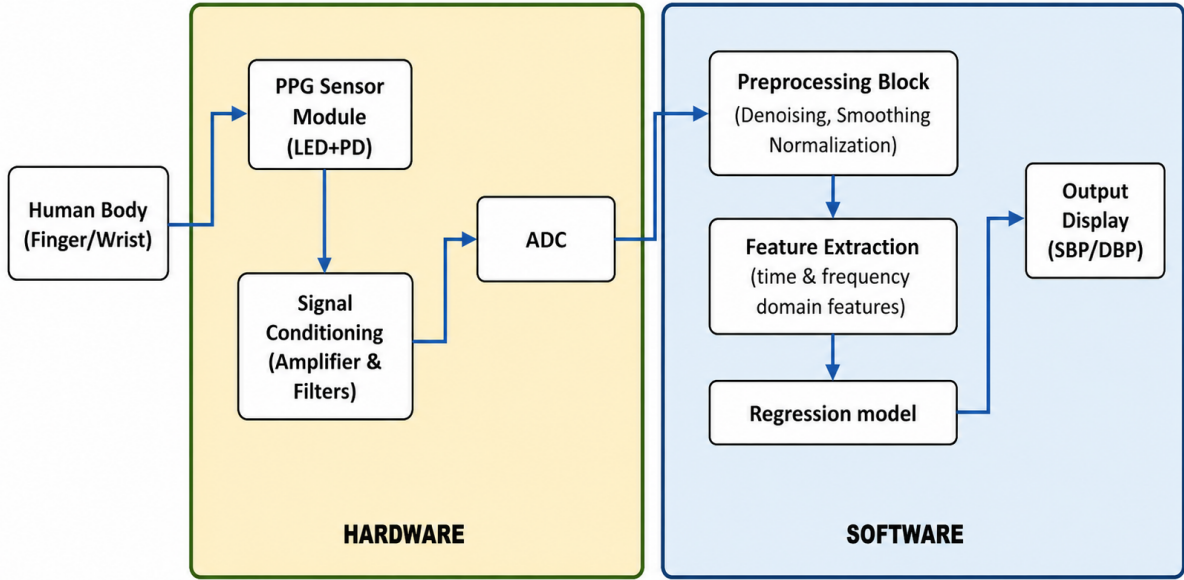


Figure 23: Proposed Workflow

Continuous monitoring of blood pressure is essential for early detection and management of cardiovascular diseases. Conventional cuff-based sphygmomanometers provide intermittent measurements and are often unsuitable for continuous or beat-to-beat monitoring. This work presents the design and implementation of a machine learning assisted system for beat-to-beat blood pressure estimation using photoplethysmography (PPG) signals along with a hardware-based real-time implementation. The proposed system aims to provide a non-invasive and cuffless approach for estimating systolic blood pressure (SBP) and diastolic blood pressure (DBP).

In the first stage of the study, PPG signals obtained from the MIMIC-III Waveform Database are preprocessed to remove noise and motion artifacts. A fourth-order Butterworth filter is employed to retain the physiological frequency components of the PPG signal while suppressing high-frequency noise. The filtered signals are then segmented and processed using a Variational Autoencoder (VAE) to extract compact and informative latent features representing the underlying hemodynamic characteristics of the waveform. These latent features are subsequently used to train a Random Forest regression model to estimate SBP and DBP values.

The performance of the proposed estimation model is evaluated using statistical error metrics including Mean Absolute Error (MAE) and Standard Deviation of Error (SDE). Experimental results demonstrate that the model is capable of predicting blood pressure with acceptable accuracy for cuffless monitoring applications.

To demonstrate practical feasibility, a hardware prototype is also developed. A PPG sensor acquires real-time physiological signals from a human subject, which are converted into digital data through an Analog-to-Digital Converter (ADC). The processed data are then transmitted to a Raspberry Pi 4 Model B that performs signal preprocessing and implements the trained machine learning model to estimate SBP and DBP in real time. The proposed end-to-end system integrates biomedical signal acquisition, machine learning, and embedded processing to enable continuous non-invasive blood pressure estimation, offering potential applications in wearable health monitoring and remote patient care systems.

6.2 Challenges Faced

Photo plethysmography (PPG) is a non-invasive optical technique used to measure blood volume changes in microvascular tissue. Although it is widely used in wearable and biomedical devices, accurate acquisition of PPG signals faces several practical challenges due to physiological, environmental, and hardware-related factors.

1. **Powerline Interference:** Powerline interference is one of the most common noise sources affecting PPG signal acquisition. It arises due to electromagnetic coupling from nearby electrical power systems operating at 50 Hz (India) or 60 Hz (USA).

(a) Causes

- i. Coupling of ambient electromagnetic fields into sensor wires.
- ii. Inadequate shielding or grounding of PPG circuitry.
- iii. Long interconnecting cables acting as antennas.
- iv. Proximity of PPG devices to electrical equipment.

(b) Effects on PPG Signal

- i. Introduction of periodic sinusoidal noise.
- ii. Distortion of the true pulse waveform.
- iii. Reduced signal-to-noise ratio (SNR).
- iv. Errors in heart rate and blood pressure estimation.

(c) Mitigation Techniques

- i. Use of notch filters (50/60 Hz).
- ii. Proper grounding and shielding of circuits.
- iii. Differential signal acquisition.
- iv. Digital filtering during signal preprocessing.

2. **Motion Artifacts:** Motion artifacts are unwanted signal distortions caused by movement of the sensor or body parts during PPG acquisition. This is a major challenge, especially in wearable and ambulatory monitoring systems.

(a) Causes

- i. Finger, wrist, or body movement.
- ii. Loosening or displacement of the PPG sensor.
- iii. Changes in contact pressure between skin and sensor.

- iv. Muscle contractions.

(b) **Effects on PPG Signal**

- i. Large amplitude fluctuations masking true cardiac pulses.
- ii. Baseline wander and waveform distortion.
- iii. False peak detection.
- iv. Inaccurate heart rate and blood pressure values.

(c) **Mitigation Techniques**

- i. Mechanical stabilization of the sensor.
- ii. Adaptive filtering techniques.
- iii. Motion artifact removal using accelerometer-based reference signals.
- iv. Signal smoothing and normalization.

3. **Low Amplitude PPG Signal:** The PPG signal inherently has a low amplitude, typically in the order of microvolts to millivolts, making it highly susceptible to noise.

(a) **Causes**

- i. Poor peripheral perfusion (cold temperature, vasoconstriction).
- ii. Low blood volume at the measurement site.
- iii. Darker skin tone or thick tissue.
- iv. Improper placement of LED or photodetector.
- v. Low LED intensity or aging of components.

(b) **Effects on PPG Signal**

- i. Reduced signal-to-noise ratio.
- ii. Difficulty in detecting systolic peaks.
- iii. Increased sensitivity to noise and interference.
- iv. Reduced reliability of extracted features.

(c) **Mitigation Techniques**

- i. High-gain, low-noise amplifiers.
- ii. Optimized LED wavelength and intensity.
- iii. Proper sensor positioning.
- iv. Signal averaging and amplification.

4. **Premature Ventricular Contraction (PVC):** Premature Ventricular Contraction (PVC) is a type of cardiac arrhythmia where an early heartbeat originates from the ventricles instead of the atria. PVC significantly affects PPG signal morphology. During a PVC event, the ventricles contract prematurely, leading to incomplete ventricular filling. As a result, the stroke volume is reduced, causing a weaker or missing PPG pulse at peripheral sites.

(a) **Causes**

- i. Abnormal electrical activity in the ventricles.
- ii. Stress, electrolyte imbalance, or heart disease.

(b) **Effects on PPG Signal**

- i. Irregular pulse intervals.
- ii. Reduced or absent PPG pulse amplitude following PVC.
- iii. Distortion in pulse shape.
- iv. Errors in heart rate variability and blood pressure estimation.

(c) **Mitigation Techniques**

- i. Detection and exclusion of PVC-affected beats.
- ii. Use of arrhythmia-aware algorithms.
- iii. Integration of ECG with PPG for accurate beat classification.

7 PPG Acquisition Circuit Designing

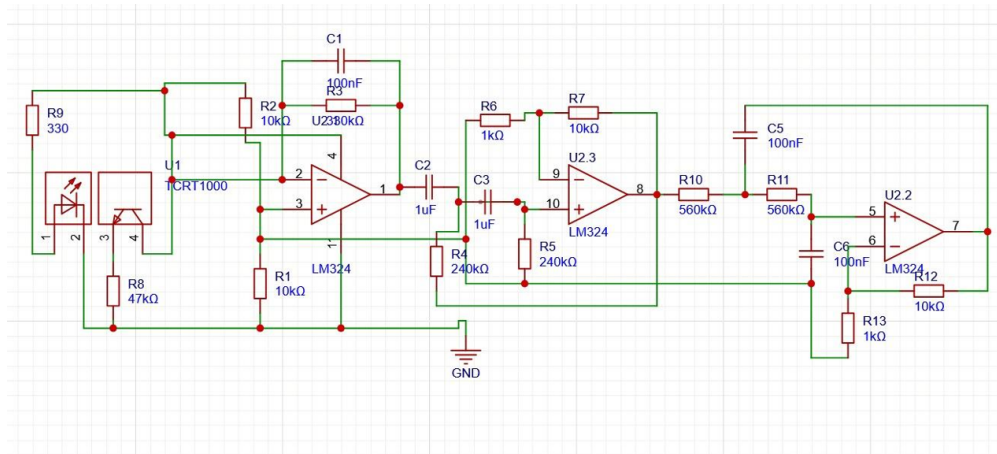


Figure 24: Signal Acquisition Circuit Diagram

7.1 Introduction

7.1.1 Background on PPG Technology

Photoplethysmography (PPG) is an optical measurement technique used to detect volumetric changes in blood in the peripheral circulation. By illuminating the skin with infrared (IR) light, the sensor measures the amount of light reflected by the tissue. Since blood absorbs more light than the surrounding tissue, the pulsatile flow of blood during each cardiac cycle results in a periodic change in the intensity of the light reaching the detector. This change generates a characteristic **PPG waveform**, from which heart rate can be derived.

7.1.2 Motivation and Problem Statement

Continuous health monitoring is critical for early diagnosis, but commercial equipment is often expensive. There is a need for **low-cost, portable** solutions. This project addresses this by using a reflective **TCRT1000** sensor. However, raw PPG signals are extremely weak (microampere range) and susceptible to noise. The main challenge is the design of an effective **Analog Front-End (AFE)** to extract a clean signal for reliable calculation.

7.1.3 Project Objectives

The primary goals of this project include:

- **Sensor Design:** Implementing a reflective-mode sensor using the TCRT1000.
- **Signal Conditioning:** Developing a TIA and active filters to eliminate noise.
- **System Integration:** Establishing a serial link between ESP32 and Raspberry Pi.
- **Local Visualization:** Interfacing an I2C LCD for immediate user feedback.

7.1.4 System Architecture Overview

The system is divided into four main functional blocks:

1. **Analog Stage:** TCRT1000 sensor + TIA + HPF + LPF.
2. **Sensing Node:** ESP32 ADC sampling and UART serial transmission.
3. **Host Controller:** Raspberry Pi for signal processing and display management.
4. **Display Interface:** 16x2 I2C LCD for real-time heart rate monitoring.

7.2 Hardware Design and Sensor Interfacing

7.2.1 The TCRT1000 Reflective Optical Sensor

The primary sensing element used in this project is the **TCRT1000**, a reflective optical sensor that includes an infrared (IR) emitter and a phototransistor in a single leaded package. Unlike transmissive sensors that require the finger to be placed between the LED and detector, the reflective sensor detects the light back-scattered from the tissue.

The emitter is biased using a current-limiting resistor to maintain a steady IR intensity. The phototransistor detects the pulsatile variations in the reflected IR light caused by the changes in blood volume in the capillary bed. The output of this phototransistor is a current in the microampere (μA) range, which requires significant amplification and conversion to be processed.

7.2.2 Trans-Impedance Amplifier (TIA)

Because the output of the TCRT1000 phototransistor is a current signal, a **Trans-Impedance Amplifier (TIA)** is employed as the first stage of the analog front-end. The TIA is a current-to-voltage converter that provides high gain while maintaining a low input impedance, which is crucial for preserving the bandwidth and linearity of the weak PPG signal.

The design utilizes an operational amplifier (Op-Amp) in an inverting configuration. The relationship between the output voltage (V_{out}) and the input phototransistor current (I_{in}) is defined by:

$$V_{out} = -I_{in} \times R_f \quad (1)$$

Where R_f is the feedback resistor. A large value for R_f (typically $1\text{M}\Omega$ to $2.2\text{M}\Omega$) is chosen to provide sufficient gain to bring the signal into the millivolt range.

7.2.3 Primary Signal Conditioning Challenges

The raw signal emerging from the TIA contains two components:

1. **DC Component:** A large offset caused by light reflected from non-pulsatile tissues (skin, bone, and muscle) and ambient light.
2. **AC Component:** The very small pulsatile signal (the actual plethysmograph) which represents only 1-2% of the total signal amplitude.

Separating these components and removing high-frequency noise from the environment (such as fluorescent lighting flicker) is the objective of the subsequent filtering stages discussed in the next section.

7.2.4 Hardware Connection: Sensor to ESP32

The output of the final analog stage is connected to the **GPIO34 (ADC1)** pin of the **ESP32**. The ESP32's 12-bit Analog-to-Digital Converter (ADC) is configured with an attenuation of 11dB to allow a full-scale input range of 0V to 3.3V, matching the output swing of the Op-Amp stages.

7.3 Signal Conditioning: Active Filter Design

The raw voltage from the Trans-Impedance Amplifier contains significant noise and a large DC offset. To extract the cardiac pulse, a dual-stage active filtering system is implemented using operational amplifiers.

7.3.1 High Pass Filter (HPF) Stage

The first stage of filtering is a **First-Order Active High Pass Filter**. Its primary purpose is to block the DC component and very low-frequency wander caused by respiration or minor body movements (the "baseline drift").

Since the typical human heart rate ranges from 40 to 200 Beats Per Minute (BPM), the fundamental frequency is between 0.1 Hz and 10 Hz. The cutoff frequency (f_c) for the HPF is typically set around **0.1 Hz**. This allows the pulsatile AC signal to pass through while centering the waveform around the reference voltage. The cutoff frequency is determined by:

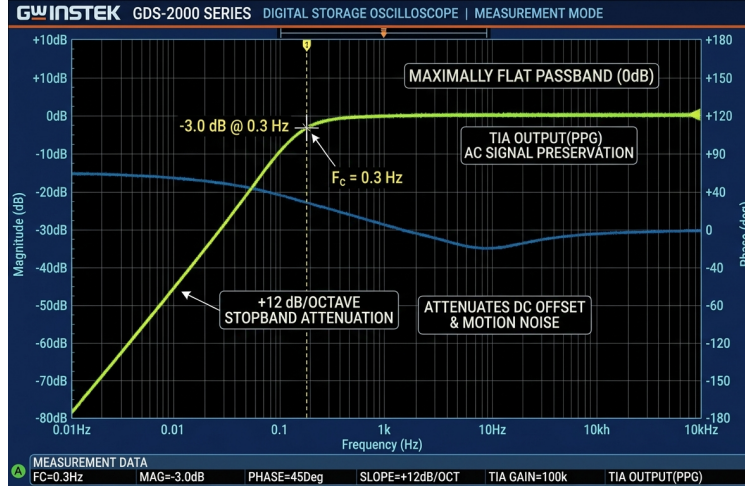


Figure 25: Output of HPF

$$f_{c(high)} = \frac{1}{2\pi RC} \quad (2)$$

By removing the DC offset, we can apply further gain to the AC signal without saturating the Op-Amp rails.

7.3.2 Low Pass Filter (LPF) Stage

The second stage is a **Second-Order Active Low Pass Filter** (2nd order Butterworth). This stage is critical for removing high-frequency interference, such as 50/60 Hz power line hum and muscle artifacts (electromyography noise).

The cutoff frequency for the LPF is set to approximately **10 Hz**. This range is sufficient to capture the main PPG peak and the diastolic notch while suppressing high-frequency noise. A second-order filter is preferred over a first-order one because it provides a steeper "roll-off" (40 dB/decade), ensuring a much cleaner signal for the ESP32's Analog-to-Digital Converter.

7.3.3 Final Stage Amplification

After filtering, the signal amplitude is often still too low for precise ADC sampling. A non-inverting amplifier stage is integrated into the filter chain to provide a final gain (typically between

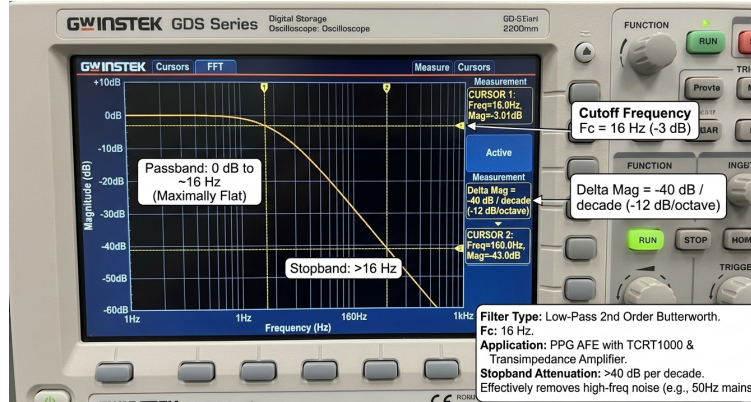


Figure 26: Output of LPF

10x and 50x). This ensures the PPG waveform spans a significant portion of the ESP32's 0V to 3.3V ADC range, improving the signal-to-quantization-noise ratio.

7.3.4 Component Selection and Stability

To ensure the stability of the analog front-end, low-noise precision Op-Amps (such as the **MCP6002** or **LM358**) are used. Decoupling capacitors ($0.1\mu\text{F}$) are placed close to the power pins of the Op-Amps to minimize power supply noise. The resulting output is a clean, rhythmic PPG waveform ready for digital processing.

7.4 Data Acquisition and Firmware Implementation

Once the PPG signal has been conditioned by the analog front-end, it is processed by the **ESP32 microcontroller**. This section details the firmware logic for sampling the waveform and the communication protocol used to interface with the Raspberry Pi.

7.4.1 ADC Configuration and Sampling Theory

The filtered analog signal is connected to one of the ESP32's 12-bit **Successive Approximation Register (SAR) ADCs**. To ensure an accurate digital representation of the PPG waveform, the sampling rate is a critical parameter.

According to the **Nyquist-Shannon Sampling Theorem**, the sampling frequency must be at least twice the highest frequency component of the signal. Given that our low-pass filter cuts off at 10 Hz, a sampling rate of **100 Hz** (10ms interval) is implemented. This rate is high enough to capture the morphology of the PPG wave (including the dicrotic notch) while remaining low enough to avoid overwhelming the serial buffer.

7.4.2 Digital Pre-processing

Although the hardware filters remove most noise, a simple digital **Moving Average Filter** is implemented in the ESP32 firmware to further smooth the signal. This helps in reducing jitter during peak detection. The algorithm maintains a window of the last five samples and calculates their mean before transmission:

$$y[n] = \frac{1}{M} \sum_{i=0}^{M-1} x[n-i] \quad (3)$$

Where x is the raw ADC value and $M = 5$.

7.4.3 UART Serial Communication

The ESP32 acts as a peripheral sensing node that streams data to the Raspberry Pi. A standard **UART (Universal Asynchronous Receiver-Transmitter)** protocol is used.

The configuration parameters are set as follows:

- **Baud Rate:** 115,200 bps (to ensure low latency).
- **Data Format:** 8-N-1 (8 data bits, no parity, 1 stop bit).
- **Packet Structure:** Each data point is sent as a string terminated by a newline character (`\n`) to simplify parsing on the Raspberry Pi side.

7.4.4 Firmware Workflow

The ESP32 software follows a non-blocking loop structure:

1. Read the voltage from the ADC pin.
2. Apply the digital moving average filter.
3. Map the 12-bit value (0-4095) if necessary for scaling.
4. Transmit the value over Serial2 (TX2/RX2 pins).
5. Wait for the remainder of the 10ms sampling period using a hardware timer.

7.5 Host Processing and Raspberry Pi Integration

The Raspberry Pi serves as the primary computational hub for the system. While the ESP32 handles high-speed data acquisition, the Raspberry Pi is responsible for signal analysis, Heart Rate (BPM) calculation, and managing the peripheral display.

7.5.1 Serial Interface and Data Parsing

The Raspberry Pi communicates with the ESP32 via its onboard hardware UART pins (GPIO 14/15) or through a USB-Serial interface. A Python-based script utilizes the `pySerial` library to poll the incoming buffer.

To ensure synchronization, the script looks for the newline character sent by the ESP32. Each incoming string is cast into an integer and stored in a sliding window array (buffer) for real-time processing. This decoupling of sensing (ESP32) and processing (Pi) ensures that the display updates do not interfere with the ADC sampling precision.

7.5.2 Data Logging and Multi-threading

The software on the Raspberry Pi is designed using a multi-threaded approach. One thread is dedicated exclusively to reading the serial port to prevent data loss, while a second thread performs the mathematical calculations and updates the user interface. This ensures that the system remains responsive even during complex calculation cycles.

7.5.3 Operating System and Environment

The system runs on **Raspberry Pi OS (Lite)**, which minimizes background CPU usage, ensuring that the timing-sensitive peak detection algorithm remains accurate. Python 3 is chosen for its extensive library support for both serial communication and I2C interfacing.

7.6 Visual Interface: I2C LCD Integration

To provide real-time feedback without the need for an external monitor, the system incorporates a **16x2 Character LCD**. This section describes the hardware interfacing and the software logic required to drive the display via the Raspberry Pi.

7.6.1 I2C Communication Protocol

A standard 16x2 LCD requires a minimum of six GPIO pins for data and control signals. To optimize the Raspberry Pi's pin availability, an **I2C (Inter-Integrated Circuit) Serial Interface adapter** (based on the PCF8574 expander) is utilized. This reduces the physical connection to only four pins:

- **VCC:** Connected to the Raspberry Pi 5V rail.
- **GND:** Common ground.

- **SDA (Serial Data):** Connected to GPIO 2 (SDA).
- **SCL (Serial Clock):** Connected to GPIO 3 (SCL).

The I2C bus operates at a default frequency of 100 kHz, which is more than sufficient for character-based updates.

7.6.2 LCD Command Set and Initialization

The Raspberry Pi interacts with the display using the `smbus2` or `RPLCD` library in Python. Upon system startup, an initialization sequence is performed:

1. **Function Set:** Configures the display for 4-bit mode and 2-line display.
2. **Display Control:** Turns on the display and enables the backlight.
3. **Entry Mode:** Sets the cursor to move to the right after every character.

7.6.3 Display Update Logic

The BPM (Beats Per Minute) value calculated by the processing thread is formatted as a string and sent to the display. To ensure the display is readable and does not flicker, the refresh rate is decoupled from the heart rate sampling rate. The display is updated every 1.5 seconds using a secondary display-refresh timer.

7.6.4 User Interface Layout

The LCD is formatted to show the system status and the final output clearly:

- **Line 1:** Displays the static label "SBP : . . DBP : . .".
- **Line 2:** Displays the live value (e.g., "Condition") followed by a heart-beat indicator. A custom character (a small box or heart) is displayed momentarily whenever a peak is detected by the algorithm, providing the user with visual confirmation of successful sensing.

7.6.5 Handling Display Latency

I2C is significantly slower than the Raspberry Pi's CPU clock. To prevent the display updates from blocking the serial data acquisition from the ESP32, the LCD functions are called within a separate low-priority thread. This ensures that the time-critical peak detection continues to run in the background with microsecond precision.

7.7 System Integration and Assembly

This section describes the physical and logical integration of the four primary modules: the TCRT1000 Analog Front-End, the ESP32 sensing node, the Raspberry Pi host, and the I2C LCD. Successful integration depends on common grounding, voltage level compatibility, and robust physical connections.

7.7.1 Physical Wiring and Interconnectivity

The system is integrated using a combination of a custom PCB for the analog stage and jumper-wire connections for the digital modules. The wiring follows a hierarchical structure:

- **Analog to Digital:** The output of the final filtering stage (Op-Amp) is connected to the ESP32's ADC1_CH6 (GPIO 34).
- **Serial Link:** The ESP32 is connected to the Raspberry Pi via USB port to serially send the sampled data. A common ground is established between both boards to ensure a stable reference for signal transmission.
- **Display Link:** The Raspberry Pi's hardware I2C pins are connected to the LCD backpack.

7.7.2 Voltage Level Shifting and Compatibility

A critical consideration during integration was the voltage difference between components. The Raspberry Pi and ESP32 both operate on **3.3V logic**. However, the I2C LCD typically requires **5V** for the backlight and character contrast. To prevent damage to the Raspberry Pi's GPIO pins, the LCD backpack is powered by the Pi's 5V rail, but the I2C pull-up resistors are tied to the 3.3V line. This ensures that the logic levels remain within the safe operating range of the Pi's Broadcom SoC while providing sufficient power to the display.

7.7.3 Power Management Strategy

To minimize electrical noise in the sensitive PPG signal, a bifurcated power strategy was implemented:

1. **Digital Power:** The Raspberry Pi is powered by a 5V/3A USB-C supply. The ESP32 is powered via the Pi's USB port, which also provides the serial-to-USB interface.

2. **Analog Power:** The Op-Amps in the TIA and Filter stages require a stable, ripple-free voltage. A dedicated linear regulator (L7805 or similar) with bypass capacitors is used to provide a clean 5V supply to the analog front-end, preventing high-frequency switching noise from the Pi's CPU from polluting the PPG waveform.

7.7.4 Ground Loop Prevention

In high-gain systems like a PPG amplifier (where gain can exceed 1000x), ground loops can introduce significant 50Hz hum. Integration was performed using a **Star Grounding** technique, where all ground references (Analog, ESP32, and Pi) meet at a single point. This prevents return currents from the digital modules from creating voltage drops across the analog ground plane.

7.8 Results and Performance Analysis

After the successful integration of the hardware and software components, the system was tested to evaluate its accuracy, signal-to-noise ratio, and reliability under different conditions.

7.8.1 Waveform Morphology Observation

The output of the analog front-end was observed using an oscilloscope and the Serial Plotter tool. The resulting **PPG waveform** displayed a clear periodic structure. Each cycle consisted of:

- **Systolic Upstroke:** A sharp rise in voltage corresponding to the heart's ventricular contraction and the arrival of the pressure wave.
- **Diastolic Decay:** A gradual fall in voltage as the blood pressure decreases.
- **Dicrotic Notch:** A small secondary peak visible in the waveform, representing the closure of the aortic valve.

The presence of a visible dicrotic notch indicates that the **Low Pass Filter (LPF)** cutoff was correctly tuned to preserve the physiological details of the signal while removing high-frequency noise.

7.8.2 Accuracy Validation

To validate the heart rate calculation, the system's output on the **I2C LCD** was compared against a medical-grade commercial pulse oximeter. Tests were conducted on five different subjects at rest.

The average error was found to be within ± 2 BPM. The minor discrepancies observed were attributed to:

1. **Quantization Error:** Limitations in the ESP32's 12-bit ADC resolution at very low amplitudes.
2. **Algorithm Latency:** The time window used by the Raspberry Pi's averaging algorithm to stabilize the BPM reading.

7.8.3 Noise and Interference Testing

The system was subjected to various lighting conditions to test the **TCRT1000's** robustness.

- **Ambient Light:** The combination of the opaque sensor housing and the **High Pass Filter (HPF)** effectively neutralized DC offsets from indoor lighting.
- **Motion Artifacts:** Significant noise was observed when the finger was moved. This is a known limitation of reflective PPG sensors, and while the digital moving average filter reduced the impact, steady contact remains necessary for high accuracy.

7.8.4 Power Consumption and Stability

The integrated system maintained stability over a 2-hour continuous monitoring test. The Raspberry Pi CPU temperature remained within normal operating limits (below 55°C), and the serial communication between the ESP32 and the Pi showed zero packet loss at the 115,200 baud rate.

7.9 Conclusion and Future Scope

7.9.1 Conclusion

The design and integration of this real-time PPG monitoring system have successfully demonstrated the synergy between analog signal conditioning and modern digital processing. By utilizing the **TCRT1000** reflective sensor, the project addressed the fundamental challenges of optical heart rate sensing, specifically the extraction of microvolt-level AC signals from large DC offsets.

The multi-stage approach—consisting of a **Trans-impedance Amplifier**, active filtering, and dual-processor integration—resulted in a stable, reliable platform for cardiovascular monitoring. The **ESP32** proved to be an efficient node for high-speed data acquisition, while the **Raspberry Pi** provided the necessary computational power for peak detection and the management of the **I2C**



Figure 27: Final Device

display interface. This architecture highlights the feasibility of building low-cost, open-source medical instrumentation that maintains a high degree of accuracy.

7.9.2 Project Limitations

While the system is functional, certain limitations were identified during the testing phase:

- **Motion Sensitivity:** Like most reflective PPG sensors, the system is highly sensitive to motion artifacts. Even minor finger movements can distort the waveform and lead to temporary BPM miscalculations.
- **Ambient IR Interference:** Although filtering is robust, extreme direct sunlight can saturate the phototransistor, exceeding the input range of the TIA stage.
- **Power Consumption:** The use of a Raspberry Pi makes the current prototype less ideal for battery-powered, wearable applications compared to a dedicated microcontroller-only solution.

7.9.3 Future Scope

The current design serves as a foundation for several advanced clinical and technical upgrades:

1. **SpO2 Integration:** By adding a Red LED alongside the existing IR LED in the TCRT1000 housing, the system could be upgraded to measure blood oxygen saturation (SpO_2) using the "Ratio of Ratios" algorithm.
2. **Wireless Web Dashboard:** Utilizing the ESP32's Wi-Fi capabilities, the data could be streamed to a cloud-based dashboard (e.g., ThingsSpeak or a local MQTT broker) for remote patient monitoring.
3. **Advanced DSP Algorithms:** Implementing a **Fast Fourier Transform (FFT)** on the Raspberry Pi would allow for heart rate detection in the frequency domain, which is significantly more resilient to noise and motion than time-domain peak detection.
4. **Machine Learning:** The Raspberry Pi could host a lightweight neural network to detect cardiac arrhythmias or irregular heartbeats by analyzing the PPG waveform morphology in real-time.

CHAPTER 8

8 Results And Discussion

Overall Pipeline

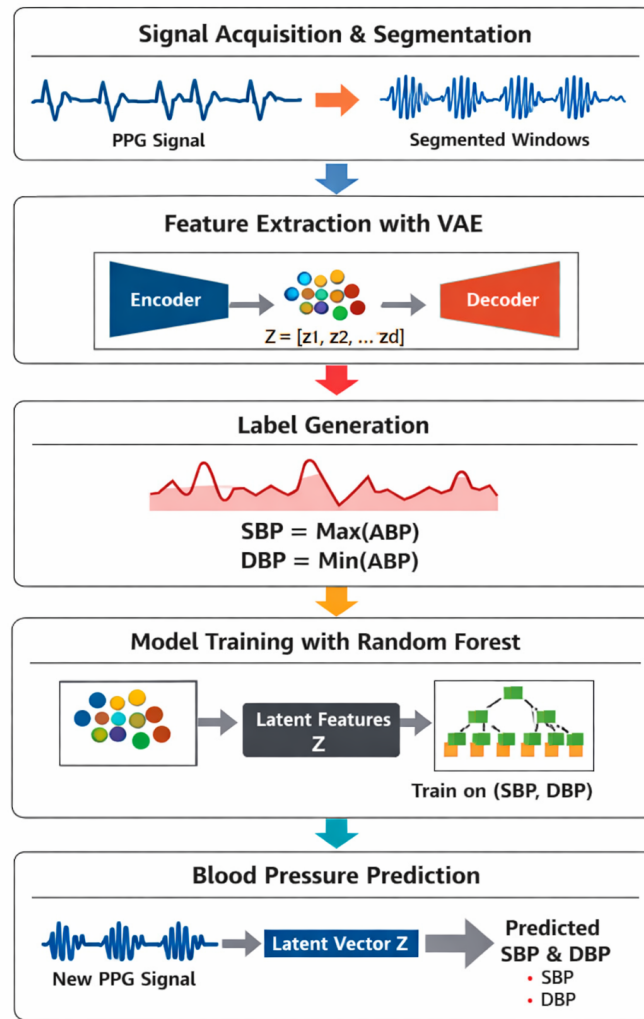


Figure 28: Flow diagram of PPG signal processing and blood pressure prediction pipeline.

The overall pipeline consists of the following steps:

1. **Signal Acquisition and Segmentation:** Continuous PPG signals obtained from the sensor are segmented into short time windows (typically 1–2 seconds) to capture beat-to-beat variations.
2. **Feature Extraction:** Each PPG segment is processed using a Variational Autoencoder (VAE), which encodes the input signal into a low-dimensional latent feature vector $\mathbf{z} = [z_1, z_2, \dots, z_d]$ representing the underlying hemodynamic characteristics.

3. **Label Generation:** The corresponding arterial blood pressure (ABP) waveform segment is used to determine the ground truth values. The systolic blood pressure (SBP) and diastolic blood pressure (DBP) are computed as:

$$\text{SBP} = \max(\text{ABP segment}), \quad \text{DBP} = \min(\text{ABP segment})$$

4. **Model Training:** The latent feature vectors $\mathbf{z}$ are used as inputs to the Random Forest regressor, while the corresponding SBP and DBP values serve as target outputs. The model learns a nonlinear mapping:

$$f : \mathbf{z} \rightarrow (\text{SBP}, \text{DBP})$$

5. **Prediction:** For unseen PPG signals, the trained VAE extracts latent features, which are then fed into the Random Forest model to estimate SBP and DBP in real time.

The performance of the model is evaluated using statistical metrics such as Mean Absolute Error (MAE) and Standard Deviation of Error (SDE), ensuring the reliability of the proposed cuffless blood pressure estimation system.

PPG Signal Preprocessing Using a 4th-Order Butterworth Filter

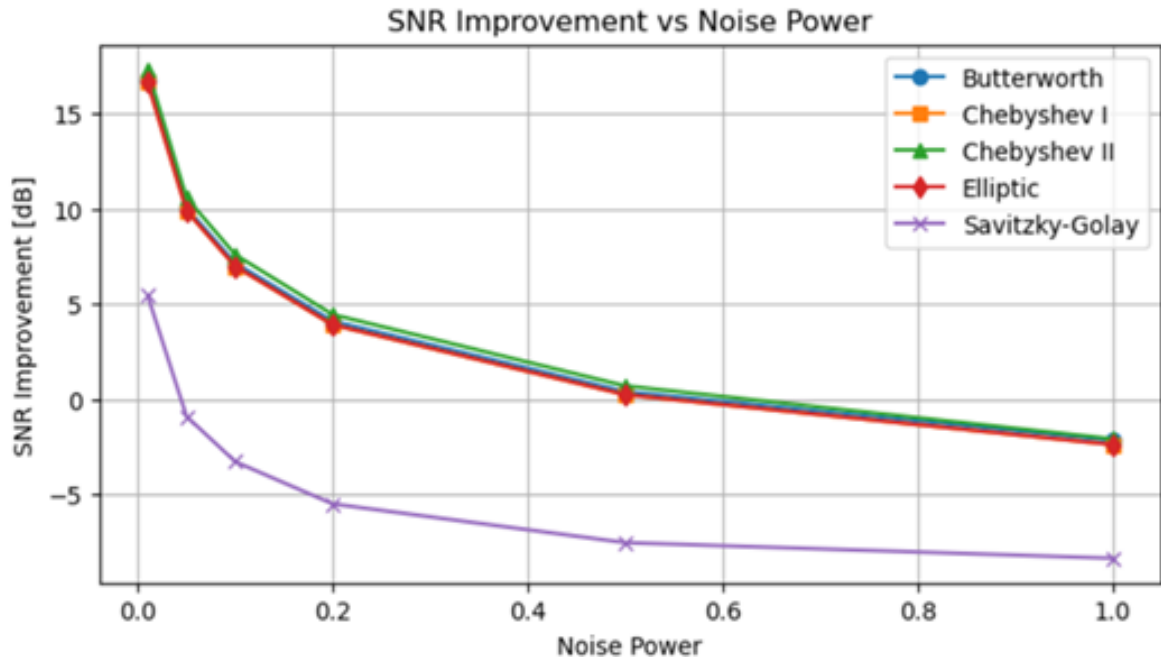


Figure 29: Graphical Representation of SNR Improvement Vs Noise Power

As we can observe in Figure 29 and in Table 7, the results show that Chebyshev type II filter works well but Butterworth filter is most favorable choice here because:

- **Maximally flat pass band response:** It means that the gain in the pass band is as close to unity as possible.

- **Monotonic roll off:** It means that the filter's response decreases smoothly and continuously as frequency increases. This provides a smooth transition between the pass band and stop band
- **Phase linearity:** It means that it does not introduce significant phase distortion into filtered signal.
- **Easy to design:** The Butterworth filter is easy to design and the cutoff frequency and the order of the filter can easily be adjusted to meet the specific requirements

Table 7: Comparison of different filters in context of SNR improvement

Filters	Response	Specification	$SNR_{i/p}$ ¹	$SNR_{o/p}$ ¹	Improved SNR ¹
Butterworth	$ H(j\omega) ^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$	Type = Bandpass, $F_s = 125$ Hz, $F_L = 0.8$ Hz, $F_H = 8$ Hz, Order= 4	16.37	-0.99	17.36
Chebyshev (Type I)	$ H(j\omega) = \frac{1}{\sqrt{1 + \varepsilon^2 T_n^2\left(\frac{\omega}{\omega_0}\right)}}$	Type = Bandpass, $F_s = 125$ Hz, $R_p = 1$, $F_L = 0.8$ Hz, $F_H = 8$ Hz, Order= 4	16.37	-0.50	16.87
Chebyshev (Type II)	$ H(j\omega) = \frac{1}{\sqrt{1 + \frac{1}{\varepsilon^2 T_n^2\left(\frac{\omega_0}{\omega}\right)}}$	Type = Bandpass, $F_s = 125$ Hz, $R_s = 40$, $F_L = 0.8$ Hz, $F_H = 8$ Hz, Order= 4	16.37	-2.75	19.12
Elliptical	$ H(j\omega) = \frac{1}{\sqrt{1 + \varepsilon^2 R_n^2\left(\frac{\omega}{\omega_c}, k\right)}}$	Type = Bandpass, $F_s = 125$ Hz, $R_p = 1$, $R_s = 40$, $F_L = 0.8$ Hz, $F_H = 8$ Hz, Order= 4	16.37	-0.56	16.93
Savitzky– Golay	$H(e^{j\omega}) = \sum_{k=-M}^M h_k e^{-j\omega k}$	Window size = 21, polyorder = 2	16.37	12.20	4.18

Latent Feature Extraction from PPG Signals Using Variational Autoencoder

Feature extraction refers to the process of transforming raw PPG signals into a set of meaningful numerical representations that capture the physiological characteristics of the waveform. These features may include time-domain, frequency-domain, statistical,

¹All performance values are measured in dB.

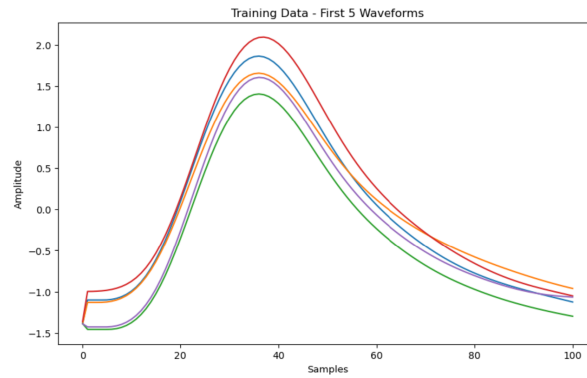
and morphological properties. In this work, instead of manually extracting features, a Variational Autoencoder is used to learn compact latent representations of the PPG signal, which are then used for blood pressure estimation.

Blood Pressure Estimation Using Random Forest Regression

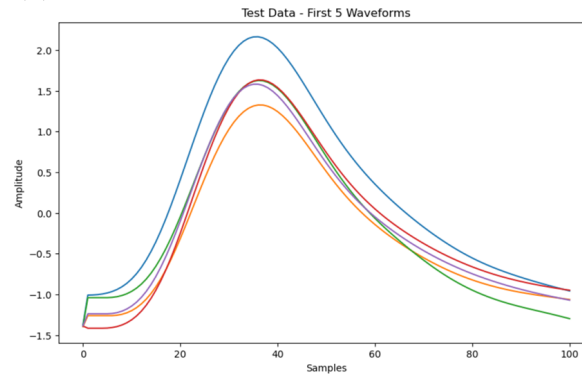
In the proposed system, Random Forest regression is employed to estimate systolic blood pressure (SBP) and diastolic blood pressure (DBP) from features extracted from photoplethysmography (PPG) signals.

Random Forest is particularly suitable for this application due to its ability to model complex nonlinear relationships, robustness to noise in physiological signals, and resistance to overfitting. Furthermore, it requires minimal parameter tuning and provides reliable performance even with moderate-sized biomedical datasets.

Performance Evaluation Using MAE and SDE



(a) Output Waveform From Training Dataset



(b) Output Waveform From Testing Dataset

Figure 30: Output Waveform For Training and Testing From MIMIC Dataset

Let y_i be the true value and $\hat{y}_i$ be the predicted value for $i = 1, 2, \dots, N$.

Mean Error (ME)

$$\text{ME} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i) \quad (21)$$

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (22)$$

Note: Mean error and mean absolute error are not same. The mean error is the average of error with the signs i.e. positive and negative signs are kept it shows whether your model is biased i.e. overestimates or underestimates. The mean absolute error is the average of absolute error (ignores the sign) it shows actual magnitude of error. Hence ME shows systematic shift and MAE shows actual accuracy.

Standard Deviation of Error (SDE)

$$\text{SDE} = \sqrt{\frac{1}{N} \sum_{i=1}^N [(y_i - \hat{y}_i) - \text{ME}]^2} \quad (23)$$

Table 8: Statistical Performance Metrics for Blood Pressure Estimation

Approach	Parameter	SBP	DBP
SMART-BP: SEMResNet and Auto Regressor based two-stage framework for non-invasive BP measurement [2]	ME	0.01 mmHg	-0.01 mmHg
	MAE	2.52 mmHg	1.47 mmHg
	SDE	3.26 mmHg	2.22 mmHg
Cuffless Blood Pressure Measurement Using Smartwatches: A Large-Scale Validation [7]	ME	2.31 mmHg	1.33 mmHg
	MAE	7.38 mmHg	5.13 mmHg
	SDE	9.57 mmHg	6.43 mmHg
Proposed Model	ME	-0.02 mmHg	-0.02 mmHg
	MAE	15.67 mmHg	7.91 mmHg
	SDE	19.72 mmHg	9.82 mmHg

Non-Invasive Blood Pressure (NIBP) standards are internationally accepted guidelines that ensure the accuracy and reliability of blood pressure measurement devices without invasive methods. They define validation procedures and performance requirements by comparing device readings with reference methods. Organizations such as AAMI, ISO, and ESH specify acceptable limits, **typically a mean error within ± 5 mmHg and a standard deviation within 8 mmHg.**

System Evaluation**1. Successful System Integration:**

The project demonstrates effective integration of analog signal conditioning with modern digital processing for real-time PPG monitoring.

2. Efficient Optical Sensing:

The use of the TCRT1000 reflective sensor enabled reliable heart rate sensing by addressing challenges such as extraction of microvolt-level AC signals from large DC offsets.

3. Robust Signal Conditioning:

A multi-stage analog design consisting of a trans-impedance amplifier, active filtering, and amplification ensured stable and noise-reduced signal acquisition.

4. Dual-Processor Architecture:

The ESP32 was effectively utilized for high-speed data acquisition, while the Raspberry Pi handled signal processing, peak detection, and I2C display control.

5. Reliable Performance:

The overall system achieved stable and consistent cardiovascular signal monitoring suitable for real-time applications.

6. Low-Cost Medical Instrumentation:

The design highlights the feasibility of developing affordable, open-source healthcare devices without significantly compromising accuracy.

Limitations of the Proposed Model

1. High Variability in Predictions

Although the Mean Error is close to zero, the Standard Deviation is high, indicating large fluctuations in predictions.

Reason: The model is not consistently capturing the relationship between PPG signals and blood pressure.

2. Latent Features Not BP-Specific

The Variational Autoencoder (VAE) is trained for signal reconstruction, not for blood pressure estimation.

Reason: Extracted features may not represent physiologically relevant BP information.

3. Lack of Physiological Feature Extraction

The model uses raw waveform data without extracting meaningful features such as peak amplitude, pulse width, or slope.

Reason: Important cardiovascular indicators are ignored.

4. Model Limitation (Random Forest)

Random Forest does not effectively capture temporal dependencies in PPG signals.

Reason: PPG is a time-series signal, and tree-based models are not ideal for sequence learning.

5. Possible Data Quality Issues

Presence of noise, motion artifacts, or imbalance in BP values may degrade model performance.

Reason: Poor-quality input leads to unreliable predictions.

To learn more about the code and to view a demonstration video of the model, please visit the GitHub repository.⁵

⁵<https://github.com/SwapnilSaha84/Design-of-ML-assisted-beat-to-beat-BP-estimator>

Future Scope

The current system provides a strong foundation for further enhancements in both clinical and technological aspects. The following improvements can be considered in future work:

1. **Reduction of Prediction Variability:** Future work may focus on reducing prediction variability (Standard Deviation Error, SDE) by training the model on larger and more diverse datasets beyond MIMIC-III, including real-world ambulatory data.
2. **Advanced Deep Learning Architectures:** Incorporating advanced deep learning models such as Long Short-Term Memory (LSTM) and Transformer networks can help capture temporal dependencies in PPG signals more effectively.
3. **IoT and Cloud Integration:** Integration with IoT and cloud platforms can enable remote patient monitoring, secure data storage, and real-time alerts for abnormal blood pressure levels.
4. **Power Optimization:** Power-efficient design techniques can be implemented to make the system suitable for long-term battery-operated wearable applications.
5. **SpO₂ Measurement Integration:** By adding a Red LED alongside the existing IR LED in the TCRT1000 sensor setup, the system can be extended to measure blood oxygen saturation (SpO₂) using the “Ratio of Ratios” method.
6. **Wireless Web Dashboard:** Utilizing the ESP32’s Wi-Fi capability, physiological data can be transmitted to a cloud-based dashboard (e.g., ThingSpeak or MQTT-based systems) for real-time remote monitoring.
7. **Advanced Digital Signal Processing:** Implementation of Fast Fourier Transform (FFT) on platforms like Raspberry Pi can enable frequency-domain heart rate detection, which is more robust against noise and motion artifacts compared to time-domain methods.



Figure 31: The Final Product

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